The Diversification of Baccalaureate Education

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Abstract

This paper identifies a decades-long trend in 4-year postsecondary education in the United States—the production of bachelor's degrees has diversified significantly and consistently by field of study. I document this pattern in multiple data sources and show that secular within-college expansion of majors drives this trend. Using a peer-network instrumental variables strategy, I find peer institution effects offer the most consistent explanation for this trend, outperforming alternatives such as changes in employer demand or state funding, shifts in student composition, or spillovers from graduate education. This peer-driven diversification is associated with higher instructional costs. New programs enter at a premium above a school's existing portfolio, one that is larger for new programs introduced under high competitive pressure from peers. This premium shows no evidence of fading and costs at existing departments are largely unaffected. These results identify peer-driven curricular diversification as one concrete, previously unmeasured channel contributing to increases in college costs.

Keywords: Higher Education, Major Choice, Competition, Peer Effects, Costs
JEL Codes: I21, I23, J24, L25, L31, C31

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1. Introduction

Students' major choices shape economic and social outcomes long after enrollment (Altonji et al., 2016; Patnaik et al., 2021), so how colleges respond to competition from other colleges can carry important consequences for the cost, accessibility, and earnings potential of human capital investments. Competition does not have a single predictable effect on what a firm produces, and for colleges the applicable case is not obvious. Does an undergraduate college sell one product—a four-year experience ending in a credential and a shared network? If so, competition should push schools to distinguish themselves from other colleges, the way single-product firms differentiate from rivals to soften price competition and carve out a defensible niche (d'Aspremont et al., 1979; Shaked & Sutton, 1982). Or does it sell many products—one per major, each with its own competitors, job prospects, and bundle of skills? If so, competition might instead push schools to match their rivals field by field, much as multi-product firms have been shown to match rather than differentiate from their competitors (Brander & Eaton, 1984). The two cases make different predictions for how a college should respond when a close competitor changes what it offers.

The market for baccalaureate education has grown increasingly stratified by institutional resources over time (Clotfelter, 2017), with competition concentrated among schools occupying similar rungs of that resource ladder. This narrowing of the competitive set has intensified as prospective students' geographic boundaries have expanded (Hoxby, 2009) and state support for public higher education has declined, pushing colleges to compete more directly for the same pool of students within their tier. Colleges commonly formalize this competitive set by identifying and tracking a specific list of peer institutions that share similar missions, goals, and constraints.

In this paper, I ask whether four-year colleges have specialized or diversified their major offerings over time, why, and at what cost. I uncover a series of new stylized facts, beginning with a decades-long trend unifying several observations about college behavior, costs, and educational quality—the production of bachelor's degrees measured by their concentration across majors has diversified significantly over time. This fact covers several narrower observations about baccalaureate education including drastic enrollment declines in the humanities (Hearn & Belasco, 2015) and growing preferences for vocational majors with clearer career prospects (Clotfelter, 2017; Getz & Siegfried, 1991). These observations alone are unable to concisely summarize overall trends in bachelor's degree production across fields. In contrast, the concept of degree diversity (i.e., the inverse of concentration) captures a new and comprehensive dimension of major choice not previously studied in the United States (US) context.[1]

Second, I show the increase in major diversity between 1990 and 2019 was due mainly to within-college expansion of major options. Large increases in the number of students obtaining a degree and changes in the racial, ethnic, and gender profiles of recipients cannot account for the trend. Instead, over half the increase can be attributed to increases in the number of majors offered by colleges. Furthermore, nearly all colleges, regardless of size, control (e.g., public or private), or highest degree, expanded their major offerings throughout the period of interest, creating less horizontal differentiation in the market for majors available to students across colleges.

Peer institutions drive this pattern, but not through the price competition emphasized in the traditional economics literature. Colleges compete over relative standing and prestige, and that kind of rivalry rewards mimicking a peer's offerings rather than contrasting them (Weisbrod et al., 2008; Marginson, 2006). Peer effect estimates using excluded peers to instrument for the average change in major diversity among an institution's chosen peers suggest an elasticity of response of at least one. That is, institutions increased their own major diversity at least in proportion to average changes among their peers. This strong effect is consistent with schools looking to attract students and improve their own quality in response to peer or aspirant institutions making similar improvements. Even though colleges tend to compete in different sub-markets (e.g., by selectivity or geography), a common response to these pressures was to expand their degree production horizontally into more fields, rather than to specialize in a few majors. This is in stark contrast to increasing market concentration, firm specialization, and declining industry diversity characterizing much of the U.S. economy's private sector (Autor et al., 2020; Ekerdt & Wu, 2023).

Building from this portrait of major diversification, I find that major diversification is also associated with higher average instructional costs. Rising costs can at least partly be attributed to the way institutions expanded into new fields, challenging explanations that emphasize colleges' inflexible production technology and reliance on high labor costs (Archibald & Feldman, 2010), or the Cost Disease (Baumol, 2012). My results suggest colleges made growth- or quality-oriented curricular decisions that drove up average costs over time. Using department-level data from The Cost Study at the University of Delaware (TCS), I show these costs accumulate because new programs enter in fields that cost roughly 1.66 times what the introducing school's existing portfolio costs on average, a premium that shows no evidence of fading as the program matures. Already-established

[1] Teixeira et al. (2013) studies this concept in Portuguese colleges highlighting prominent differences between private and publicly controlled institutions. As I will show, this is quite distinct from the U.S. case where diversification is nearly universal in the time-period of interest.

departments within schools show limited evidence of cost disruption when new programs are introduced. Furthermore, the new-program cost premium is significantly larger for those introduced under high competitive pressure from peers, directly tying these cost consequences to the peer mechanism described herein.

With these results I contribute new context for analyses of colleges and educational choices. Other studies in this vein have used differential tuition prices by major (Andrews & Stange, 2019; Stange, 2015), GPA restrictions for specific majors (e.g., Andrews et al., 2017; Bleemer & Mehta, 2021, 2022), and changes in enrollment capacity (Bianchi, 2020; Thomas, 2024) to identify changes in student behavior and outcomes. Cook (2021) explicitly modeled the number of majors a college offered arguing this had a place in colleges' objective functions through its costs. Doing so suggested students were willing to pay about $50 for an additional program. I build on this work focusing more broadly on major diversity and on the motivations for and consequences of adding programs.

More broadly, this paper speaks to a literature on peer effects among organizations rather than individuals. Most empirical work on peer effects concerns people, students, workers, or neighbors, but a smaller literature shows firms imitate one another too, for instance in capital structure and payout decisions (Leary & Roberts, 2014). Within higher education specifically, colleges' own decisions, from admissions policy to pandemic reopening plans, are known to track those of their peers (Blair & Smetters, 2021; Acton et al., 2022). Nonprofit organizations may be especially prone to this kind of imitation. Without a traditional profit motive, it has long been argued they compete on quality and prestige rather than price (e.g., Newhouse, 1970), which gives peer standing an outsized role in shaping their behavior. I provide some of the most direct evidence yet of this kind of peer effect in a higher-education setting, using an instrument that isolates plausibly exogenous variation in peers' behavior and estimating an elasticity of response of at least one.

I also provide new insight to the literature on costs and the market structure of higher education. Several explanations have been offered for rising costs (for a review of this literature see Cheslock et al., 2016). The Cost Disease view diminishes colleges' role in the process, arguing that costly labor cannot easily be substituted or made more efficient in educational production (Archibald & Feldman, 2010; Baumol, 2012). Other explanations place more onus on institutional responses to incentives, arguing that colleges engage in a sort of arms race for more resources and prestige, sometimes attributed to a lack of concrete connections between spending and student outcomes (e.g., Blair & Smetters, 2021; Bowen, 1980; Clotfelter, 1996). A similar dynamic among hospitals, where non-price competition drives a costly "medical arms race," has been well documented (Kessler & McClellan, 2000; Gaynor et al., 2015). This paper provides suggestive evidence for this account in higher education. Major diversification, as it reflects competitive responses to what a college's peers are doing, is also associated with higher instructional costs.

This paper proceeds in distinct parts, peeling back sequential layers of the major diversification phenomenon. In sections 2 and 3, I first identify the trend in major diversity across the 4-year sector of higher education and briefly describe the data used to decipher it in detail for the rest of the paper. In section 4, I show this trend is widespread across most colleges and attributable to within-college changes to program offerings, particularly growth in the number of majors a school offers, a channel I return to throughout the paper. In section 5 I argue for and show peer effects to be the strongest explanation for this behavior, ruling out other alternatives. In section 6 I analyze the consequences of major diversification on instructional costs, documenting their positive relationship and ties to program introduction and "peer pressure." I conclude the paper with a discussion of implications and directions for future research in section 7.

2. Trends in major diversity

To measure diversity of college majors over time I turn to two large-scale public data sources, the American Community Survey (ACS) and Integrated Postsecondary Education Data System (IPEDS). Beginning in 2009, the ACS began collecting college major from students with a bachelor's degree. Pooling survey years from 2009 to 2019, I generate cohorts roughly representing the year respondents would have graduated college, assuming college exit at age 22. The large sample in the ACS allows me to track majors for graduating cohorts back to 1970.[2] Following Hemelt et al. (2023) and Conzelmann et al. (2023), I collapse the 173 ACS major codes into 69 categories for comparability across data sources and more stable groupings over time. Each category is best thought of as a department within a college and avoids concerns of capturing changes due to simpler repackaging of existing courses into new "programs," instead favoring more structural changes to curricular offerings.

As a second data source, I turn to IPEDS. This federally mandated survey of all colleges and universities in the US contains degree award records dating back to the late 1980s. I begin my panel in 1990 and measure bachelor's degrees awarded through 2019. The raw IPEDS data are reported

[2]Technically, I could measure cohorts back as far as the 1930s and 40s, but the sample sizes become very small. I choose 1970 because it is a rounded year and one for which there is adequate (weighted) sample of respondents.

at the 6-digit Classification of Instructional Programs (CIP) code level, which I collapse to the 69 categories using the crosswalk developed in Hemelt et al. (2023) in order to facilitate comparison to the ACS.[3]

Throughout the paper, I will measure major diversity using what I call the Effective Major Index (EMI). This is defined as the inverse of a Herfindahl-Hirschman Index (HHI), a common measure of industry or employment concentration. I index majors, $m \in M$, where M is a the set of 69 majors. The EMI in any given year t can be written as,

$$\text{EMI}_t = \frac{1}{\text{HHI}_t} = \left[\sum_{m \in M} \left(\frac{x_{mt}}{x_t} \right)^2 \right]^{-1}, \tag{1}$$

where x_t is the total number of degrees awarded in year t and x_{mt} is the number of individuals who obtained a degree in major, m in that year. The EMI captures the diversity of bachelor's degrees awarded in a given year. An EMI of 1 would imply all students in that year majored in the same field. The more dispersed students are across different fields, the EMI increases with a ceiling of 69. This would imply equality of degrees across all possible fields within a given year where each major would have a share of 1/69.

Figure 1 plots the diversity of college majors across all graduates calculated with the ACS from 1970 to 2017 and with IPEDS from 1990 to 2019. Though the data generating processes for these sources are quite distinct,[4] they both illustrate a decisive trend: students have diversified their majors significantly over time.

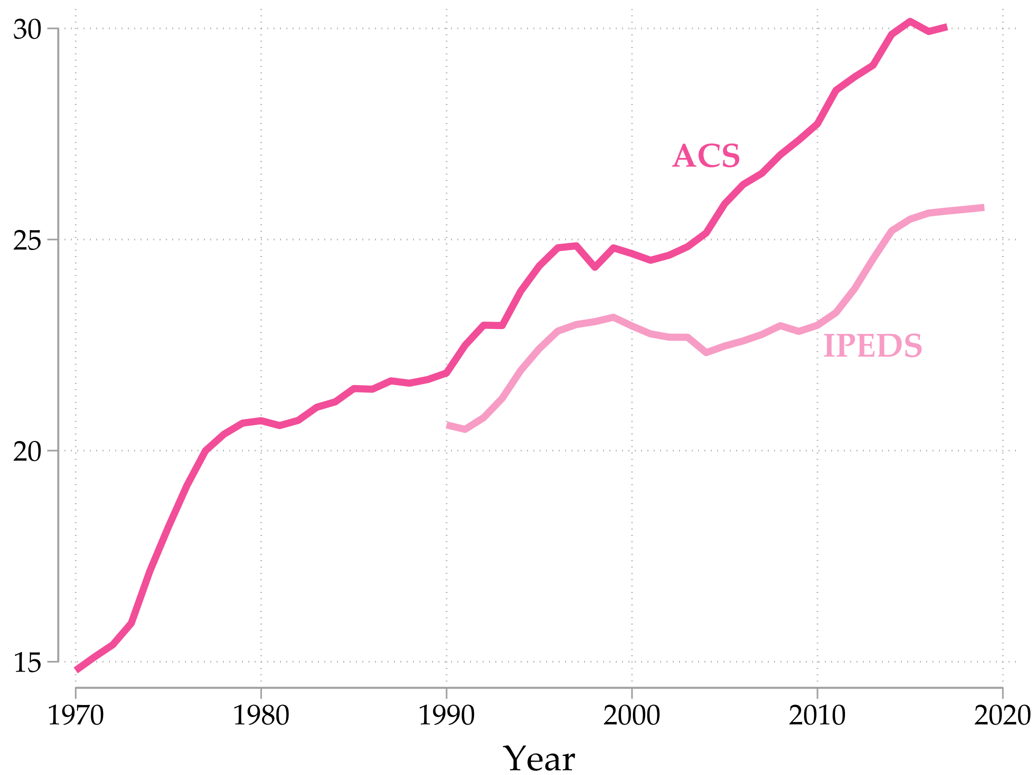
The upward trajectory is particularly stark in the 1970s and 1990s from the ACS. This pattern is mirrored in the 1990s by the IPEDS data as well. The two sources diverge somewhat in the 2000s, with IPEDS flattening, until the two sources trend steeply upward together again for much of the 2010s. The numbers themselves imply that for the last three decades, the effective number of majors for bachelor's degrees have risen between 20 and 30 percent or about 5 to 7 effective majors. This implies the dispersion of students across majors has become significantly more even over time, instead of heavily favoring a few larger majors as was the case in earlier years of the panel.[5]

[3]Three different vintages of 6-digit CIP codes were used to track degree awards in this time period (versions 1990, 2000, and 2010), which I harmonized at the 4-digit level prior to collapsing in the final 69-category major classifications used in the paper. More information about this process can be found in the Data Appendix B

[4]ACS contains self-reported education levels and majors while IPEDS degree counts are reported by colleges and universities.

[5]The trends in major diversification described here are robust to alternative measurement choices. Reconstructing EMI at coarser (2-digit CIP) or finer (4-digit CIP) aggregation than the 69-category scheme used throughout yields a similar

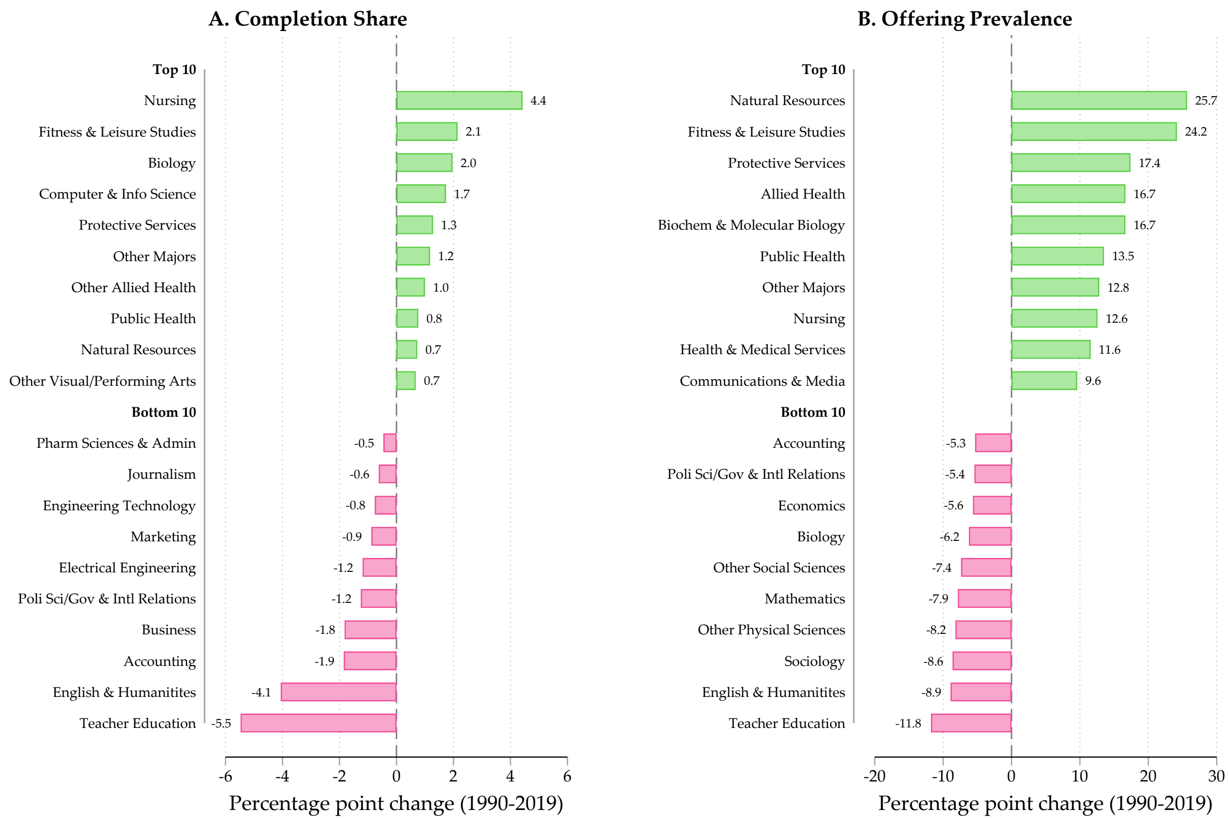
Figure 1: Effective Major Index from the ACS (1970-2017) and IPEDS (1990-2019)



Notes: The EMI is calculated as the inverse of the sum of squared shares of 69 possible bachelor's degree majors in each year. See Equation 1.

The fields with the largest gains and losses in completion share from IPEDS between 1990 and 2019 are depicted in Figure 2. Nursing gained more ground than any major, increasing its share by 4.4 percentage points. Education dropped the most at 5.5 percentage points. These two alone cloud the suggestion that students and colleges have come to favor "vocational" majors to more general degrees over time. Teaching is among the most specific majors for its direct path to a particular occupation, and yet its share has fallen dramatically. Similar juxtapositions can be made in other common gaining or declining majors like Accounting, which lost almost 2 percentage points, versus Computer Science, which gained about 1.7 percentage points.

Figure 2: Top and bottom fields by change in completion share and offering prevalence, 1990–2019



Notes: Each panel displays the ten fields with the largest increases (Top 10) and the ten fields with the largest decreases (Bottom 10) on the relevant measure over the period 1990 to 2019, among colleges in the analytic sample. *Panel A* shows the change in each field's share of total bachelor's degrees awarded, expressed in percentage points. *Panel B* shows the change in offering prevalence—the fraction of sample colleges awarding at least one bachelor's degree in that field—also expressed in percentage points. Pink bars indicate decreases; green bars indicate increases. Fields are ranked separately within each panel. "Other Majors" comprises CIP 4-digit codes not classified into any named field, including military science and ROTC programs (CIP 28–29), vocational and trades programs (CIP 46–49), and small niche interdisciplinary specializations such as Medieval and Renaissance Studies and Peace Studies.

Changes in offering prevalence, or the share of colleges awarding degrees in a particular field,

upward trajectory, as does down-weighting or excluding second-major completions in the post-2001 subsample for which IPEDS separately identifies them. See Appendix Table B1.

tell a complementary story. Between 1990 and 2019, new program offerings outnumber exits by roughly four to one across fields (16,380 new offerings against 4,096 exits, comparing 1990 to 2019 at the school-by-field level).[6] Most individual fields changed modestly in prevalence, but growth concentrated in a tail of fields that moved from rare to fairly common, such as Public Health (offered by 1 percent of colleges in 1990 to nearly 15 percent by 2019) and Natural Resources (6 percent to 32 percent). Appendix Table A1 contains the full set of field level changes.

Looking across majors, whether by their share of degrees or how widely they are offered, provides only mixed support for previous attempts to categorize broad trends in 4-year degrees. Several vocational majors have either increased or decreased in share. Humanities degrees have fallen in both measures, but so have some business-adjacent fields, like Accounting and Economics. Narrow comparisons fail to capture how major choices and production have evolved over time. The EMI captures this more thoroughly by measuring diversity. It is agnostic to major type (e.g., vocational vs. general) and reflects the number and dispersion of majors offered rather than the trajectory of a small set of fields.

3. Data sources

To explore the origins of major diversification and its ramifications for higher education, I recruit the use of several data sources. In general, the years of interest for this study run from 1990 through 2019, covering three full decades of information. I briefly document the main sources in the subsections that follow, though more robust descriptions of sample selection, harmonization, and validity checks can be found in Appendix B.

3.1. IPEDS

The majority of the data for this paper come from IPEDS. What can be described as a mandatory census of all colleges and universities that receive federal funds for higher education, IPEDS is both comprehensive and relatively complete, at least with respect to basic measures of institutions in the US. I limit my sample to schools and years between 1990 and 2019 in which at least one bachelor's degree was awarded. This includes public, private non-profit, and for-profit colleges in all 50 states and Washington, D.C., but I exclude schools in Puerto Rico and other outlying islands and territories. The sample includes some schools traditionally viewed as 2-year institutions (e.g., community colleges), but only in years when they may have awarded a bachelor's degree. The

[6] This is a pure endpoint comparison; it does not capture fields a school dropped and later re-added within the panel.

sample contains 3,236 total institutions between 1990 and 2019. This number grows significantly over time, from 1,724 institutions in 1990 to 2,333 in 2019. 1,423 institutions awarded degrees during all 30 years of the panel and the mean number of years degrees were awarded overall was about 20.

Outside of degree completions, I harmonize and include institution characteristics, like control, highest degree offering, a school's location, and enrollment and degree completions by race/ethnicity and gender. I also obtain yearly full-time equivalent (FTE) enrollment of undergraduate and graduate students,[7] and the state of residence for incoming first-year students.

I also harmonized institution-level expenditures and revenues to explore how major diversification affects costs per student. I focus on instructional expenditures and the total financial support generated from state governments. I adjust all expenditure and revenue variables using the Consumer Price Index (CPI) to 2019 dollars for analysis, unless otherwise noted.[8]

The final data elements I use from IPEDS are school-identified lists of peer institutions. The peer lists provide institutions a way to compare themselves to others along core metrics reported to IPEDS, like enrollment, completion, financial aid, tuition, etc. The comparisons are made in a public *Data Feedback Report*, which is an annual report of the institution's own data along with the mean of the peer institutions it identified in the reporting process.[9] I provide more description of peer lists and discuss their analytic use in Section 5.

3.2. The Cost Study at the University of Delaware

In some analyses of costs, I use department-level data collected as part of TCS. Conducted yearly with data available between 1998 and the present, TCS is a voluntary data collection of department-level direct instructional costs, credit hours taught, and faculty teaching loads. Schools opt to participate in the study yearly and need not participate every year. I subset the full sample to schools that reported instructional expenditures and student credit hours that were also in my IPEDS analytic sample for an unbalanced panel of 622 institutions. I subset this list to 97 "perennial participants," or those who have ten or more consecutive years of data, for a more consistent and stable panel (see Appendix B for more details). This perennial sample skews toward larger, public, doctoral-granting institutions relative to the full IPEDS analytic sample; Appendix Table B2

[7]For details on estimating this for earlier years see Appendix B

[8]I also use the Higher Education Price Index (HEPI), which is another measure of inflation arguably more specific to the types of costs typically incurred by higher education institutions. My results using this method of adjustment do not substantively change the results.

[9]See here for an example: https://nces.ed.gov/ipeds/DFR/2019/ReportHTML.aspx?unitid=199120

compares the two directly.

These data are typically used by reporting institutions to benchmark their own instructional spending against others in the sample. Recently, the data have been used to explain differences in instructional costs per student credit hour across different fields (Hemelt et al., 2021) and to estimate the responsiveness of credits and faculty distributions to shocks for major-specific employer demand for skills (Conzelmann et al., 2023).

While TCS sample is not necessarily representative of all 4-year colleges in the US, I show in appendix Figure A1 the EMI in terms of credits earned across participating institutions maps very closely to the bachelor's degree EMI from IPEDS between 1998 and 2019.

4. The Origins of Major Diversification

To distinguish different factors that may have led to major diversification I begin by generating college-level EMI values over time. The college-year EMI can be expressed,

$$\mathrm{EMI}_{it} = \left[\sum_{m \in M_i} \left(\frac{x_{imt}}{x_{it}} \right)^2 \right]^{-1}, \tag{2}$$

where I sum the squared shares of each major m within the set of majors, M_i offered at a given school, i. This yields a value on the same scale as in equation 1 with a ceiling unique to the number of programs offered at each school in year t. Throughout the rest of the paper, my interest will be in the degrees-weighted value of EMI_{it}. This does not equate to the overall population EMI_t because schools with varying sizes may specialize (or not) in different fields to varying degrees.[10]

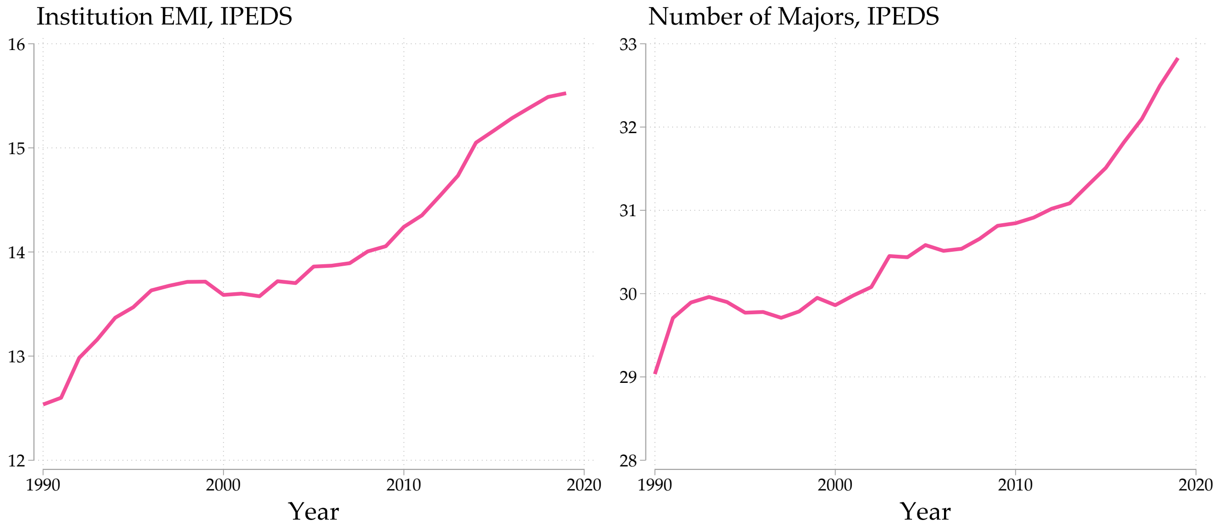
I plot the weighted EMI in Figure 3 on the left-hand side. And on the right, for comparison, I plot the raw weighted average number of majors in which degrees were awarded. The average number of programs is also a measure of degree diversity but disregards the distribution of awards across fields. By visual inspection, the trends in the EMI map closely with an increase in the average number of majors within schools. This relationship need not be mechanical and is entirely possible for major diversity to increase holding the number of majors in each school fixed.[11] If this were the case, the average number of majors over time would remain relatively flat, suggesting students shifted their choices across a relatively fixed set of major offerings. However, the trend in major

[10] Consider an extreme scenario where all schools award degrees in a single field. If we allow school size to vary, the population EMI will vary over time based on the allocation of students across schools, while the within-school EMI would remain 1 for all years, by construction.

[11] For instance, a hypothetical school with just two majors has students who prefer major one 3:1 in time period 1, yielding an EMI of 1.6. In period 2, students shift to parity, 2:2, yielding an EMI of 2.

offerings is positive and highly correlated with the EMI.

Figure 3: Institution EMI and number of majors, 1990-2019



Notes: Both sets of estimates are weighted by degrees granted. EMI=effective major index. The total number of programs come from the full list of 69, as referenced in the main text.

4.1. Diversification was primarily within colleges, but also homogenized the market

To understand what is driving the overall increase in the EMI, I decompose year-over-year changes in institution-level EMI into a within-college component, capturing colleges adding majors, and a between-college component, capturing the changing mix of institutions producing bachelor's degrees, following Melitz & Polanec (2015) and Olley & Pakes (1996). First, I define the $\log(\text{EMI}_{it}) = Y_{it}$. The degrees-weighted average log EMI across all colleges in year t can be written,

$$Y_t = \sum_{i \in I} \left(\frac{x_{it}}{x_t} \right) Y_{it}, \tag{3}$$

where the term in parentheses is the share of all bachelor's degrees awarded in year t at a given institution i. I decompose the year-over-year changes to Y_t where institutions can fall in and out of the sample in each period. For any two years of the panel, $t = 1, 2$, institutions are classified into one of three groups of *continuers*, new *entrants*, and *exiters*. These groups are indexed by $g \in \{C, E, X\}$ respectively. Continuing institutions are those that award degrees in both periods. Entering schools awarded degrees in the second period, but not the first. And schools categorized as exiters awarded degrees in the first period, but did not award any in the second. I decompose the change in Y_t from $t = 1$ to $t = 2$ into:

$$\Delta_1^2 Y = \underbrace{\left(\frac{x_2^E}{x_2}\right)\left(Y_2^E - Y_2^C\right) - \left(\frac{x_1^X}{x_1}\right)\left(Y_1^X - Y_1^C\right)}_{\text{net entry}} + \underbrace{\Delta\overline{Y}^C}_{\text{within } C} + \underbrace{\Delta\text{cov}_C\left(\frac{x_i}{\overline{x}^C}, Y_i\right)}_{\text{between } C} \tag{4}$$

Here, x_t^g refers to the number of degrees awarded in group g and Y_t^g is the weighted average log EMI for each group in the respective time periods 1 or 2. The first bracketed expression quantifies the contribution of entering and exiting colleges on the change in log EMI based on their share of the total degrees and major diversity relative to continuing institutions. A positive value for net entry implies that the degrees awarded by schools who entered during the period were more diverse relative to continuing and exiting colleges, where a negative value would imply the opposite.

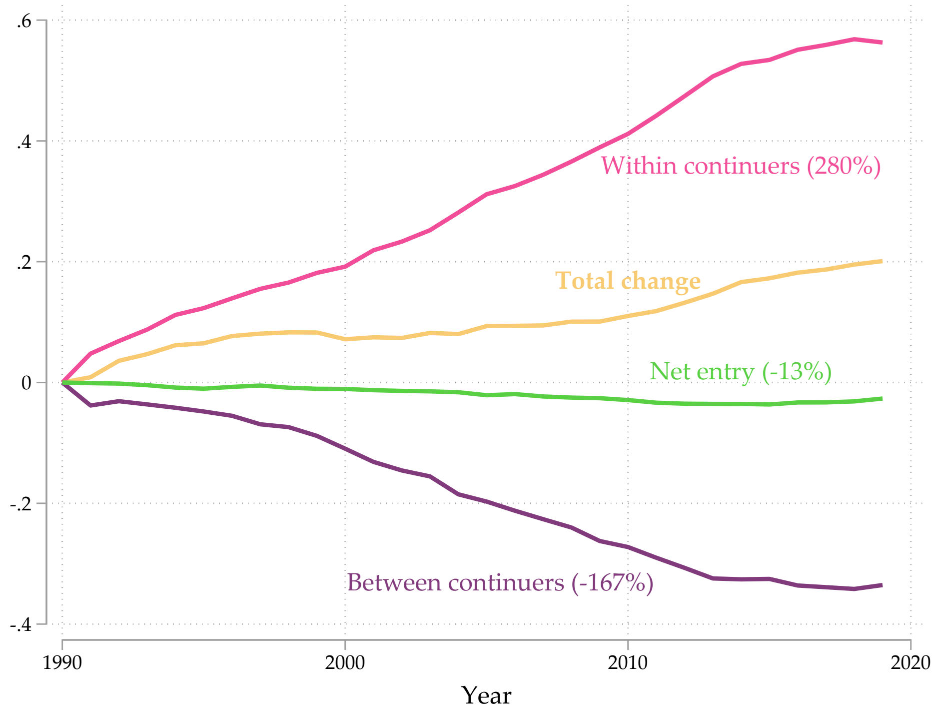
The second bracketed term captures the change in log EMI attributable to changes within continuing colleges. Here, $\overline{Y}^C$ is the average unweighted change in the log EMI among continuing colleges. Positive values imply the average continuing college diversified their degrees awarded, where negative values imply declining diversity. Large values in the positive (or negative) direction suggests a more ubiquitous trend across institutions becoming more (less) diversified in their degrees awarded.

The third and final bracketed term measures the contribution of between-college variation in the EMI among continuing schools. This term helps quantify the relationship between diversification and institutional size. It is defined as the change in the covariance between the relative size of each continuing institution, where $\overline{x}^C$ is the average size of continuing colleges, and the college-specific EMI, Y_i. Interpreting its value depends on the sign of the covariance. A declining covariance could indicate a strengthening negative relationship between college size and degree diversity ($\text{cov}_2 < \text{cov}_1 < 0$).[12] A declining between-college term could also indicate a declining positive relationship between size and diversity ($0 < \text{cov}_2 < \text{cov}_1$). This implies a pre-existing relationship between college size and major diversity that has decreased or become less prominent over time. Degree production would no longer be as predictive of major diversity in this case.

The results of this exercise depicted in Figure 4 reveal several interesting facts about the evolution of bachelor's degree diversity. The total change in the EMI from 1990 to 2019 of 0.2 log points was driven by two strong bifurcating forces. The first was a large, positive diversification

[12]This case is the focus of Ekerdt & Wu (2023) and their analysis of the decline in industry diversity of manufacturing firms in the US, where its large negative values over time suggests heterogeneity in the type of firm driving the decline in diversity. Namely, large firms moved toward specialization, arguably driven by demand for higher quality goods in the US over time.

Figure 4: Decomposition of the change in log EMI 1990-2019



Notes: This figure decomposes the total change in the log EMI into three main components described in Equation 4. The percentages sum to 100, representing the share of the total change.

of majors within continuing colleges. This within-school diversification was almost 3 times the aggregate change at 0.55 log points. The second was a steep decline in between-college variation. The relationship between college size and major diversity declined by about 0.33 log points throughout the period.

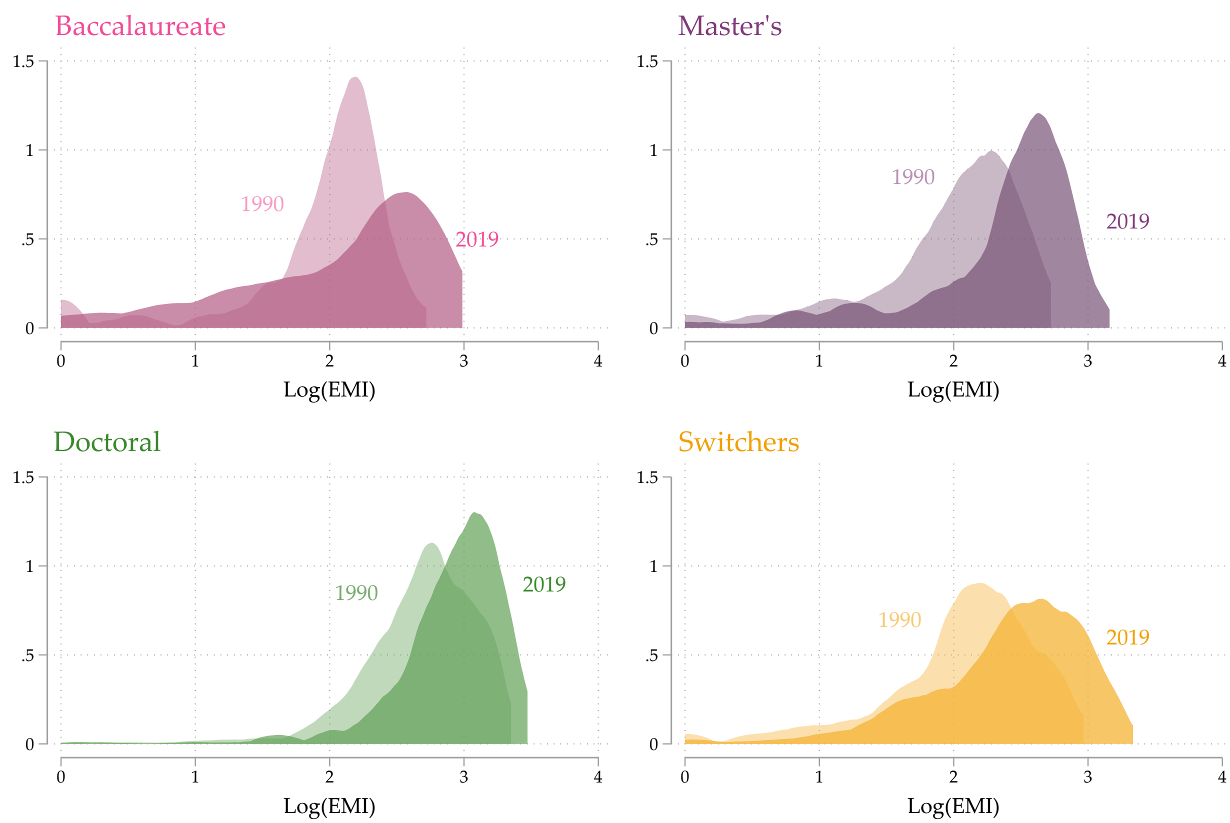
The large within-college value suggests the typical behavior of a college throughout this period was to diversify. The decline in between-college variation refines this story suggesting that smaller institutions like liberal arts and master's institutions were diversifying at a faster rate than larger comprehensive institutions who began the period with higher levels of major diversity. In effect, these colleges caught up to comprehensive research universities and looked much more like them in terms of their bachelor's degree production in 2019 than in 1990.

Net entry in the market for bachelor's degrees plays a minimal role in explaining the total change in major diversity (about -0.03 log points). Starting a new college is costly as is offering bachelor's degrees for the first time on top of other programs. Schools who enter are not likely to compete in the share of degrees with well established institutions.[13]

[13] This decomposition partially masks the role entering institutions play in the long-run. Over a quarter of institutions in the full panel "entered" the market at some point after 1990. The persistence of once-entrants contributes a large

To further illustrate the main takeaways from this exercise, I plot the distributions of the log EMI for four different types of institutions based on their highest educational offerings in 1990 and 2019. Figure 5 shows that majors for bachelor's degrees diversified, regardless of highest degree offering. This is visually apparent from the shift in each distribution rightward from 1990 to 2019. "Switchers" in the bottom right panel of Figure 5 in orange are schools that added or eliminated degree capacity in the panel.[14]

Figure 5: Density of the log EMI in 1990 and 2019, by highest degree offering



Notes: Each panel is a group of institutions based on the highest degree offered in 1990 and 2019. "Switchers" are those whose highest degree offering was different in 1990 than in 2019, dominated by schools adding higher degrees (94%).

The distribution for doctoral institutions in green shifted right less sharply, where it was more pronounced for the other groups. Doctoral institutions in 1990 tended to house more colleges and departments than smaller master's and baccalaureate institutions. Yet, since 1990 their share of bachelor's degrees awarded declined 5 percentage points (from 55 to 50 percent) as many other in-

portion to the overall trend, as they became continuing colleges after initial entry. This decomposition is shown in appendix Figure A2.

[14]Switching anytime between 1990 and 2019 is quite prevalent, encompassing nearly 40 percent of institutions and 34 percent of all degrees by the end of the panel in 2019. The vast majority of schools in this group gained capacity (94%) adding higher degree offerings over time, rather than eliminating them.

stitution types expanded their degree-granting capacity and added major options along the way.[15]

4.2. Oaxaca-Blinder Decomposition

To isolate the contribution of program growth from other explanations for major diversification, I decompose the change in the log EMI from 1990 to 2019 into observable student characteristics, institutional attributes, and a residual term using an Oaxaca-Blinder decomposition.

First, Table 1 presents the means and standard deviations for these key variables across institutions in 1990 and in 2019 along with their differences. As discussed in the previous section, the EMI increased overall by 0.2 log points in three decades. During this time, undergraduate enrollment changed dramatically both in scale and in the composition of students attending. FTE enrollment increased by over 0.35 log points and was particularly strong among URM, like Hispanic students, whose share of all degrees increased by about 10 percentage points, and Black students and women, whose share increased by about 4 and 5 percentage points respectively.[16]

Colleges also expanded their curricular offerings both for baccalaureate and graduate education significantly over this period. For instance, the share of BAs being awarded by schools offering a doctoral degree grew from 56 to over 78 percent in 2019. Another interesting descriptive fact is that institutions grew their number of BA program offerings by about 0.13 log points between 1990 and 2019. As I will show, this growth in the number of majors is a key driver of overall major diversification nationally.

I also include the total share of non-hospital revenue for each institution generated from state appropriations and state financial aid (e.g., need or merit-based grants to students). Reliance on state funding fell from 30 percent to less than 17 percent during this time.

Following other similar work to decompose changes in graduation rates over time by Bound et al. (2010) and Denning et al. (2022), I follow their two-component structure, separating the change in log EMI into a portion explained by shifts in the distribution of observable characteristics and an unexplained residual, rather than the three-part decomposition with a separate coefficients (returns) effect common in labor market applications of Oaxaca-Blinder. I perform my EMI decomposition on the full sample and different institution types separately. These sets of results can be found in Table 2. In the first column with the full sample, the chosen covariates explain about

[15] I formalize the contribution each type of college made to overall major diversification in Appendix Figure A3, which reveals that the colleges adding degree capacity (switchers), like graduate programs, were responsible for the largest share of the increase in major diversity.

[16] Unfortunately, measures of student socioeconomic status or academic preparedness upon college entry are not available in IPEDS until much later in the panel, or are only available for a subset of schools (e.g., test scores).

Table 1: Changes among Bachelor's degree granting colleges between 1990 to 2019

	Mean (1990)	SD (1990)	Mean (2019)	SD (2019)	Difference
Log(EMI)	2.377	0.618	2.578	0.668	0.201
Log(Number of BA programs)	3.235	0.628	3.362	0.637	0.126
Log(Undergraduate FTE)	8.851	1.076	9.202	1.154	0.351
Student characteristics					
Share BAs awarded Black	0.056	0.115	0.092	0.118	0.036
Share BAs awarded Hispanic	0.031	0.057	0.136	0.134	0.105
Share BAs awarded Foreign/Intl	0.025	0.033	0.052	0.056	0.027
Share BAs awarded Other non-white	0.041	0.064	0.114	0.090	0.073
Share BAs awarded Race unknown	0.030	0.114	0.034	0.051	0.004
Share BAs awarded Women	0.530	0.107	0.574	0.101	0.045
Institution attributes					
Public Highly Selective	0.057	0.232	0.099	0.299	0.042
Public Less Selective	0.601	0.490	0.560	0.497	-0.041
Private Highly Selective	0.068	0.252	0.085	0.278	0.016
Private Less Selective	0.268	0.443	0.210	0.408	-0.058
For-profit	0.006	0.075	0.046	0.210	0.040
Highest degree offering: Bachelor's	0.099	0.299	0.048	0.214	-0.051
Highest degree offering: Master's	0.343	0.475	0.170	0.375	-0.173
Highest degree offering: Doctorate	0.558	0.497	0.782	0.413	0.224
Share of degrees BA level	0.733	0.153	0.703	0.162	-0.030
State funding					
Share of revenue from state	0.304	0.229	0.167	0.156	-0.137

Notes: Intl=International. All estimates are weighted by the number of bachelor's degrees awarded by a given institution in 1990 or 2019 respectively. Selectivity is determined based on Barron's Competitiveness Index where highly selective includes the top two categories and less selective includes all others. Reliance on state funding is calculated as the share of total non-hospital revenue derived from state appropriations and financial aid to students.

two-thirds or 67 percent of the observed increase in the log EMI.

Table 2: Oaxaca-Blinder Decomposition of Change in Log(EMI) from 1990 to 2019

	Full Sample	Public highly selective	Private highly selective	Public less selective	Private less selective	For-profit
Log EMI 2019	2.578	3.042	2.606	2.761	2.164	1.208
Log EMI 1990	2.377	2.591	2.356	2.528	2.033	0.722
Total Change	0.201	0.451	0.250	0.232	0.131	0.486
Explained changes	0.134	-0.166	0.082	0.154	0.071	0.839
	(67%)	(-37%)	(33%)	(66%)	(54%)	(173%)
Student characteristics	-0.066	-0.381	-0.084	-0.058	-0.071	0.078
	(-33%)	(-84%)	(-34%)	(-25%)	(-54%)	(16%)
Institution characteristics	0.045	0.0002	-0.003	0.042	0.007	-0.067
	(22%)	(0%)	(-1%)	(18%)	(5%)	(-14%)
State financial support	0.046	-0.022	0.004	0.071	-0.004	-0.027
	(23%)	(-5%)	(2%)	(31%)	(-3%)	(-6%)
Log(Undergraduate FTE)	0.008	0.014	-0.003	0.013	0.001	0.033
	(4%)	(3%)	(-1%)	(6%)	(1%)	(7%)
Log(Number of BA Programs)	0.101	0.223	0.168	0.086	0.138	0.822
	(50%)	(49%)	(67%)	(37%)	(105%)	(169%)
Unexplained (residual)	0.067	0.616	0.168	0.078	0.060	-0.353
	(33%)	(137%)	(67%)	(34%)	(46%)	(-73%)

Notes: Student characteristics include the share of BA recipients that are Black, Hispanic, Foreign/International, of unknown race/ethnicity, all other non-white, and the share that were awarded to women. Institution characteristics include the highest degree offering (bachelor's, master's, or doctorate), the share of all degrees awarded that were BAs, and institution category combining selectivity (top-two Barron's categories and all others) with control (public or private non-profit), and a separate group for for-profit institutions. State financial support is the share of total non-hospital revenue derived from state appropriations and financial aid to students. Both undergraduate enrollment measured in FTE, and the number of BA programs were log transformed for this exercise.

Compositional changes in students obtaining bachelor's degrees negatively contributed to overall diversification, conditional on supply-side factors and state funding: absent the growth in female and non-white graduates between 1990 and 2019, I estimate major diversity would have risen another 7 percent. This is consistent with persistent gaps in STEM attainment by gender (e.g., Sloane et al., 2021) and by race (e.g., Gelbgiser & Alon, 2016), and with GPA-based enrollment restrictions in some majors that disproportionately affect URM and low-income students (Bleemer & Mehta, 2021).

Changes to overall undergraduate enrollment explains less than 5 percent of the positive change in the EMI, conditional on the other factors in the decomposition. This is consistent with the declining positive relationship between institution size and degree diversity shown in the previous section. About 22 percent of the change in EMI was related to other institutional attributes, particularly increases in the share of institutions offering master's and doctoral degrees.

Institutional reliance on state funding was associated with about a 5 percent increase in the EMI (23 percent of the total change). Lower levels of state funding seen in 2019 compared to 1990 were predictive of higher major diversity. This pattern was driven mainly by non-selective public institutions who make up a large share of degree awards and rely more heavily on state funding to operate and attract students. Among the most selective public institutions, state funding declines do not appear to have had an impact on their increases in major diversification. Across institution groups, declining revenue share from the state was accompanied by colleges diversifying their degree offerings.

The most prominent factor was the (log) number of BA programs offered: about half the aggregate increase in EMI is attributable to growth in program counts, a relationship that holds across institution types and is the single strongest predictor throughout.[17]

This descriptive evidence points to a prominent role for within-institution diversification of BA programs in explaining the rise in the EMI. In contrast, compositional changes in students obtaining a bachelor's degree drove down major diversification. As the market for bachelor's degrees in the US swelled, colleges and universities grew to look more like one another, choosing to expand into new undergraduate programs and less so vertically within their established programs.

5. Peer institutions and Major Diversification

In this section I show that major diversification can be attributed to competition among peer institutions in the market for bachelor's degrees. Most colleges cannot distribute profits, and even those that can operate under substantial constraints on price-based competition from financial aid regulation and reputational concerns. Competing on quality is a primary tool for attracting students and resources, and major diversity is one dimension of that competition. Specifically, colleges behave in ways consistent with improving their own relative quality or prestige among a group of close peer institutions. This explanation outperforms others including changes to employer demand and the business cycle, state funding for higher education, and spillovers from graduate education.

[17] Log(Number of BA Programs) and the EMI are correlated but not mechanically identical; a school's EMI can rise while holding its major count fixed, as illustrated in the example accompanying Figure 3 above. The explained share attributed to program counts exceeds 100 percent of the total change in some smaller subsamples (e.g., private less-selective and for-profit institutions), reflecting the added noise of decomposing within smaller, more heterogeneous groups rather than a mechanical relationship between the two measures.

5.1. Defining Peer Institutions

To define peer institutions, $p(i)$, I use lists that colleges submit as part of IPEDS reporting and public *Data Feedback Reports*, which schools use to benchmark their own performance against the peers they name. These lists can be updated yearly but are remarkably stable over time, with average year-over-year overlap within an institution between 0.95 and 0.99. Lists submitted a full decade apart overlap by about 0.77 (See Appendix Table A2). This high level of stability suggests that schools only marginally change who they consider to be their peer institutions over time.

I further provide descriptive information on how institutions list peers in Table 3. On average, colleges tend to choose "aspirant" peers. A school's own 6-year graduation rate and instructional expenditures per FTE are about 90 and 94 percent that of the peer group average across all years of data, and schools tend to list peers with higher enrollment than they themselves possess. The same pattern holds for selectivity: the average institution stands slightly below its peer group, with a mean selectivity-standing index of -0.19 on a scale from -1 to $+1$, using Barron's Competitiveness Index value comparisons.

Table 3: Focal Institution to Peer Group Comparisons and Network Characteristics

Variable	Mean	p10	p25	p50	p75	p90	N
Focal-to-peer comparisons							
Instructional expenditures per FTE ratio	0.942	0.551	0.707	0.869	1.050	1.320	42,773
Six-year graduation rate ratio	0.903	0.622	0.781	0.911	1.011	1.116	29,556
Undergraduate FTE ratio	0.794	0.200	0.503	0.758	1.012	1.262	43,171
Selectivity standing	−0.186	−0.824	−0.545	−0.167	0.143	0.438	35,771
Network characteristics							
Peer-list popularity	24.14	12.71	17.36	21.72	28.17	33.58	43,265
Distance (miles)	583.15	117.95	250.22	509.42	795.62	1122.17	43,265
Reciprocity	0.191	0.000	0.053	0.154	0.292	0.444	43,348
Clustering	0.202	0.056	0.089	0.165	0.267	0.417	43,286

Notes: All statistics are computed across school-year observations. Focal-to-peer ratios—instructional expenditures per FTE, the six-year graduation rate, and undergraduate FTE—are calculated as the focal institution's value divided by its peer group average in a given year; instructional expenditures per FTE are CPI-adjusted. Selectivity standing is the mean, across a focal institution's peers, of a signed indicator equal to $+1$ when the focal institution is more selective than the peer, -1 when less selective, and 0 at the same Barron's Competitiveness Index value, so that positive values indicate standing above the peer group; it is therefore a signed index ranging from -1 to $+1$. Peer-list popularity is the number of times an institution is named as a peer by other institutions. Distance is the average crow-flies distance in miles between the focal institution and each of its peers. Reciprocity is the share of a focal institution's listed peers that also list the focal institution in return. Clustering is the share of a focal institution's peer pairs that are also peers with each other. Peer-list popularity, reciprocity, and clustering are fixed characteristics computed from the static peer lists described in Section 5, observed across varying numbers of school-years as institutions enter and exit the analytic sample. Columns beginning with "p" refer to percentiles. N reports school-year observations.

The same table also reports the network's structural characteristics. The geographic spread of peer groups is quite vast where the average distance from a focal institution to its peers is 583 miles, but 10 percent of colleges choose average peers within about 118 miles while another 10 percent list peers over 1,100 miles away on average. Schools are named as peers about 24 times each on average, and the networks show modest reciprocity and clustering: roughly a fifth of a school's listed peers name it in return, and a fifth of its peer pairs also list one another. I return to these network measures in Section 5.5, where I test whether the estimated peer effect is concentrated in particular kinds of networks.

Peer-list submission to IPEDS is optional, and lists have only been collected since 2010, so coverage of the analytic sample is not universal. I present in Table 4 basic descriptive statistics for institutions who did and did not submit lists. Schools that submitted a list at least twice account for 85 percent of all bachelor's degrees awarded in my sample between 1990 and 2019, despite coming from 53 percent of institutions. Schools that never submit are disproportionately for-profit or non-selective in admissions (31 vs. 16 percent and 87 vs. 44 percent) and award degrees for fewer years of the panel (15 vs. 25 years on average) suggesting schools that submit peer lists are more established, more likely to be engaging in some selective admissions process, and much larger producers of the country's bachelor's degrees overall.

I subset each school's cumulative set of peers from all possible lists to the top-10 most submitted institutions, including ties, and require that a school be listed at least two times to be considered a close peer. This improves the overall stability of the lists across time. I assign one static list of peers for each institution throughout the panel, varying only if peers drop in or out of being a bachelor's degree producer (i.e., not in the analytic sample in a given year). This implicitly assumes peer groups identified in the last decade of the panel were also peers in the earlier two decades as well. Section 5.5 presents a test of this assumption, restricting estimation to the 2010–2019 window in which the network is actually observed (see Appendix Table A7) along with several other network sensitivity checks.

5.2. Peer Effects Empirical Strategy

To estimate how colleges change their major diversity in response to their peers, I begin with a canonical linear in means specification from the peer effects literature (e.g., Manski, 1993; Moffit, 2001).

Table 4: Descriptive statistics of institutions that select peers versus those that do not

	Did not Select Peers		Selected Peers	
	Mean	SD	Mean	SD
Number of Institutions	1,517		1,719	
Share of institutions	0.47		0.53	
Share of all BA's awarded (1990-2019)	0.15		0.85	
Control: Public	0.14	0.34	0.31	0.46
Control: Private non-profit	0.56	0.50	0.53	0.50
Control: For-profit	0.31	0.46	0.16	0.36
Selectivity: High	0.01	0.12	0.10	0.30
Selectivity: Moderate	0.11	0.31	0.46	0.50
Selectivity: Low	0.87	0.33	0.44	0.50
Degree-weighted Avg Log(EMI)	2.14	0.75	2.55	0.62
Avg yearly BA's awarded	183.94	576.95	776.39	1247.97
Avg years BA's awarded (max 30)	13.53	10.81	25.22	8.41
Avg share degrees: Associate's	0.29	0.35	0.16	0.29
Avg share degrees: Bachelor's	0.55	0.33	0.64	0.28
Avg share degrees: Graduate	0.16	0.25	0.20	0.21
Highest degree offering: Bachelor's	0.53	0.50	0.24	0.42
Highest degree offering: Master's	0.26	0.44	0.33	0.47
Highest degree offering: Doctorate	0.21	0.41	0.44	0.50

Notes: Statistics are unweighted unless noted otherwise. Institutions that selected peers are those who submitted lists of peer institutions as part of IPEDS reporting and Data Feedback Reports. I include only schools who submitted these lists 2 or more times between 2010 and 2019. Selectivity is derived from Barron's Competitiveness Index where highly selective groups the first two categories (Most and Highly Competitive), moderately groups the next two (Very Competitive and Competitive) and the third category includes all others. Averages within school are taken over all possible years during which BAs were awarded according to IPEDS.

$$Y_{i,t} = \beta_1 \overline{Y}_{p(i),t} + \mathbf{\Gamma} \boldsymbol{X_{i,t}} + \mathbf{\Pi} \boldsymbol{\overline{X}_{p(i),t}} + \eta_i + \eta_t + \epsilon_{i,t}, \tag{5}$$

where Y is the the log EMI regressed onto the degrees-weighted average log EMI of an institution's peers, a vector of time-varying institution characteristics, $\boldsymbol{X_{i,t}}$, analogous time-varying average peer characteristics $\boldsymbol{X_{p(i),t}}$, and school and year fixed effects. The vector of time-varying controls captures changes to student composition that is potentially correlated to peer shocks and to changes in the focal institution's EMI. This vector includes the share of BAs awarded to Black and Hispanic students, women, and foreign students. It also includes the share of the student body enrolled part-time and the share of graduate students. To account for differential trends in the evolution of the EMI that may occur for institutions newer to the market for bachelor's degrees (e.g., former 2-year college), I also include in $\boldsymbol{\overline{X}_{i,t}}$, a school's "age" and its quadratic for the number of years it awarded BAs within the panel at each time, t. Finally, I include (log) undergraduate enrollment of the focal institution and peers' average. Even though capacity is one way institutions could respond to changes in peer quality, my interest is in isolating the effects of major diversification net of enrollment increases.[18]

Given the length of the panel, institution fixed effects may not capture meaningful "fixed" differences over time, and recent work by Millimet & Bellemare (2026) formalizes how correlated heterogeneity and dynamic misspecification can bias unit fixed-effects estimates, a concern that grows more relevant as panels lengthen. As a robustness check, I therefore re-estimate the model in stacked 6-year long differences, which removes fixed characteristics of the institution over each 6-year interval. The long-difference estimates are reported in Appendix Table A4 and are consistent with the fixed-effects results.

Identifying peer effects is difficult for two main reasons. First, the reflection problem, highlighted by Manski (1993), arises in trying to separate a group effect on an individual from the individual's effect on the group, particularly when the outcomes are determined simultaneously. The existence of institution-specific peer groups in my setting helps to alleviate this concern. For example, there are excluded peers that are peers of peer institutions that are absent an institution's own list. On the issue of simultaneity, my estimated effects are robust to lagging the peer measures by one or two years.

The second problem in identifying peer effects is the potential existence of unobserved factors

[18] It is worth noting that enrollment was not a significant predictor of the change in the EMI from Section 4.2. The results presented here are also robust to excluding controls for enrollment.

affecting a group of institutions that are also correlated with major diversification. In my setting, this could be labor market shocks affecting students' college decisions (like where and what major to pursue), which might also relate to an institution's decision to diversify its major offerings. To overcome this issue, the non-overlapping peers can be used to instrument for peer behavior. This point is illustrated by Bramoullé et al. (2009) using a hypothetical triad between a focal institution i, its peer p, and its peer k, where p affects i, and k affects p, but k only affects i through its effect on p and not directly.

Following this logic, I instrument for first degree peers' EMI $(\overline{Y}_{p(i)})$ using measures from excluded peers of peers, defined by the function $k(\cdot)$, collecting institutions that are peers of the focal institution's peers, $p(i)$, but not part of the institution's own set.[19] One intuitive approach is to instrument for first peers' EMI using the log average EMI of excluded peers, $\overline{Y}_{k(i)}$. Another approach, illustrated by De Giorgi et al. (2010), is to use average covariate values of excluded peers as instruments instead of their outcome values. In this case, a vector of average student-body characteristics for excluded peers, $\overline{\boldsymbol{X}}_{\boldsymbol{k(i)}}$, is likely to be highly correlated with the outcomes, but perhaps less susceptible to correlated effects. These overidentified models have a precedent in the peer effects literature and require a generalized two-stage least squares (2SLS) estimation procedure (GMM 2S) as in Lee (2003).

Either of these IV strategies may yet be biased from strategic listing of peer institutions. Peer lists themselves could be endogenous to the choices schools make to undergraduate curriculum and offerings, which would be problematic if schools select institutions strategically based on their expectations for how it would look for comparisons. Fortunately, schools do *not* appear to be selecting peers to improve their relative appearance, at least generally speaking. As described in the previous section, schools tend to choose "aspirant" peers with 6-year graduation rates and instructional expenditures per FTE higher than that of the focal institution. This behavior is preferable for identification since institutions should be more likely to follow the major diversity behavior of institutions at or above their own output than those below them. That aspirant institutions are well-represented in the peer lists also diminishes the concern that true peers would show up as excluded peers having been omitted from the observed list.

The IV strategy could also fall short of addressing correlated effects among peer groups without

[19] One concern is that $k(i)$ may contain schools that are in a college's genuine competitive set, but not listed by the focal institution. I address this directly in Section 5.5 by dropping schools from $k(i)$ that list the focal institution as a peer, and, more aggressively, dropping schools from $k(i)$ resembling a true peer statistically using a peer-selection model estimated on realized peer choices (see Appendix Table A8).

"network" fixed effects (Bramoullé et al., 2009). Schools might self-select into peer groups based on shared characteristics, like admissions selectivity, geography, or prestige, that are omitted or unobserved in the model yet also correlated with the evolution of major diversity. In my setting, I do not have clear network delineations barring institutions of some type from being peers of another.[20] I address this concern directly in Section 5.5, testing whether the estimated effect is concentrated in more homogeneous or tightly clustered networks, which is where correlated effects driven by shared characteristics would be expected to show up most.

5.3. Peer Effect Estimates

Table 5 presents the fixed-effects estimates under various strategies for dealing with endogeneity concerns. Across specifications, the evidence for peer effects in this market is strong. Both the ordinary least-squares (OLS) and Instrumental variables (IV) approaches show a large and positive elasticity of response to peers in the focal institution's major diversity. In column 1, OLS estimates imply that focal institutions increase their major diversity by about 3 percent for each 10 percent increase in the peers' average EMI.

The IV estimates are significantly larger than those from the OLS specifications, with elasticities ranging from 1.2–2. Columns 2, 4, and 6 using the average major diversity among peers of peers $(\overline{Y}_{k(i)})$, yield elasticities of 1.98, 2.00, and 1.67 under year, state-by-year, and linear-trend fixed effects, respectively. The year and state-by-year estimates are both statistically distinguishable from an elasticity of one ($p = 0.001$ for each one). The first stage is strong throughout, with F-statistics above 100 in the fixed-effects columns and stronger under the linear trend.[21] Columns 3, 5, and 7 instrument using average peer-of-peer covariates $(\overline{\boldsymbol{X}}_{\boldsymbol{k(i)}})$; first-stage F-statistics are just above 20 and the resulting elasticities (1.21, 1.19, and 1.36) are modestly smaller and not statistically distinguishable from one ($p = 0.442$, $p = 0.482$, and $p = 0.145$, respectively). Hansen's J fails to reject the overidentifying restrictions in any specification ($p \geq 0.37$). Neither instrument's estimate moves meaningfully across the three fixed-effects choices, suggesting the elasticity is not an artifact of how the common time trend in the EMI is removed. Appendix Table A4 presents analogous estimates using stacked 6-year long differences, confirming that the peer effect is robust to the estimator choice concern discussed above.[22]

[20] In individual peer-effect literature, networks are often bounded by things like schools or classrooms.

[21] Detailed estimates from the first-stage can be found in Appendix Table A3.

[22] The main results are also robust to how majors are classified. Reconstructing the EMI at coarser or finer CIP aggregation leaves both the aggregate trend and the peer elasticity essentially unchanged, as does down-weighting or dropping second-major completions in the post-2001 subsample where they can be separately identified (Appendix Table B1).

Table 5: Peer Institution Effects on Major Diversification

	Year FE			State×Year FE		Linear Trend	
	1	2	3	4	5	6	7
	OLS	2SLS	GMM 2S	2SLS	GMM 2S	2SLS	GMM 2S
Peers' Log(Average EMI)	0.311**	1.978**	1.207**	1.997**	1.193**	1.670**	1.363**
	(0.082)	(0.291)	(0.269)	(0.311)	(0.275)	(0.200)	(0.249)
Instrument(s)		$\overline{Y}_k$	$\overline{\boldsymbol{X}}_{\boldsymbol{k}}$	$\overline{Y}_k$	$\overline{\boldsymbol{X}}_{\boldsymbol{k}}$	$\overline{Y}_k$	$\overline{\boldsymbol{X}}_{\boldsymbol{k}}$
First-stage F		110.0	22.3	101.9	23.7	194.6	20.5
Hansen's J			6.895		7.563		5.972
(p)			0.440		0.373		0.543
N Observations	43,057	42,903	42,903	42,877	42,877	42,903	42,903
N Clusters	1,702	1,690	1,690	1,690	1,690	1,690	1,690

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All specifications include institution fixed effects. All models are weighted by the base-year total number of BA degrees awarded. Column 1 presents the traditional linear-in-means estimate including institution covariates and corresponding peer group averages. These controls include the log undergraduate enrollment, share of a school's total enrollment that is part-time, share graduate students, the share of bachelor's degrees awarded to women, to students that were either Black or Hispanic, to foreign or international students, and the number of years in the panel the school had awarded BA degrees and its quadratic. Columns 2 and 4 instrument for peers' log(EMI) using the average peers of peers' log(EMI), $\overline{Y}_k$. Columns 3 and 5 are estimated using a 2-stage generalized method of moments weighting procedure where excluded instruments are the control variable averages (X's) of all peers of peers, identified by $k(i)$, and Hansen's J tests the overidentifying restrictions. Columns 6 and 7 replace year fixed effects with a continuous linear year trend. Appendix Table A4 presents analogous estimates using stacked 6-year long differences as a robustness check addressing potential bias from time-varying institutional fixed effects (see Section 5).

Appendix Table A6 reports additional specifications including lag and weighting checks, along with fixed effects for institutional selectivity, spending, and graduation-rate tiers interacted with year, all of which provide consistent estimates of peer effect elasticities. Estimation restricting to the 2010–2019 window in which the peer network is directly observed provides similar but, in some cases, less precisely estimated elasticities given the substantially smaller sample.[23]

Regardless of specification, the IV elasticities definitively exceed the OLS elasticities, which is the opposite signature of standard peer effect concerns. OLS is often contaminated by common shocks a school shares with its peers, like a shared labor market or a state funding change, which push both outcome and treatment in the same direction. The instrument in that case is expected to pull the estimate downward. Peer groups here overlap only partially, so a shock affecting a school's peers need not extend to that school's second-degree peers in the same way, breaking the logic by which a shared shock would bias OLS (De Giorgi et al., 2010). Separately, IV and OLS need not agree even absent any bias since the two estimators weight schools differently, OLS by degree volume and 2SLS by how strongly a school's peer EMI moves with its second degree peers' EMI, and nothing requires these weighting schemes to favor the same schools or to imply the same average effect. A larger IV estimate is consistent with either channel and does not on its own indicate a problem with the instrument.

The IV estimates apply only to schools whose peers were induced into diversifying their majors by their own peers. In a separate exercise, I characterize these likely "compliers" by creating a binary treatment for peers' EMI split at its median, and instrument for it using the same median split of the peers-of-peers' instrument, $\overline{Y}_k$. The results suggest compliers skew toward moderately selective, public institutions (73 v. 64 percent and 78 v. 65 percent, respectively) with somewhat smaller enrollment and lower instructional spending than the full estimation sample (Appendix Table A5). This is consistent with middle-rung institutions responding most actively to competitive pressure from their peers.[24] The main elasticities should therefore be read as more representative of this segment of the market rather than all baccalaureate institutions.

[23] Appendix Table A7 reports the restricted-window results in full, including versions with a linear or state-specific trend that restore first-stage power.

[24] Complier characteristics are estimated using the kappa-weighting method of Abadie (2003). The propensity scores are computed using the same degrees weighting, controls, and fixed effects as the main regression specifications in column 2 of Table 5.

5.4. Peers and New Program Adoption

New programs are the largest contributor to major diversification over the period of study, and the peer effects documented above operate at least partly through this channel. If colleges are responding to peers' curricular decisions, they should be more likely to introduce a new program in a given field when their peer institutions already offer it or have been expanding degree production in that field. I test this directly using a survival model in which each major a school does not yet offer is treated as "at risk" of adoption in each year. The dependent variable takes a value of zero in all years a program is not offered and one in the year a school first begins awarding degrees in that major, at which point it exits the risk set.

Table 6 presents estimates of peer adoption on the probability of new program introduction. Each column reports a separate linear probability model estimated by OLS and, in the fully-specified model, by 2SLS instrumenting peer adoption with the average of the same measure among the school's second-degree peers, paralleling the EMI models of the previous section. Columns 1-3 use the number of peers currently offering the major as the treatment; columns 4-6 use total peer degree awards in that field scaled to hundreds, capturing the intensity of peer activity rather than just prevalence. Within each group, the first column includes major-by-year fixed effects only, which absorb common trends in field-level demand across all institutions, and the others add school-by-major fixed effects, which additionally absorb time-invariant school-field tendencies.

Colleges are more likely to introduce a program when more of their peers already offer it. One additional peer offering a major raises the probability of adoption by 0.1 percentage points, an increase of about 12 percent from the baseline adoption rate of 0.9 percent. The effect is larger when conditioning on school-by-major fixed effects where one additional peer raises the adoption probability by 0.3 percentage points, a 41 percent increase from baseline. The peer degree production measure tells a similar story, with effects ranging from 7 to 24 percent of the baseline across OLS specifications. Instrumenting peer adoption with peers-of-peers roughly doubles the estimates, to 79 and 47 percent of baseline for the two measures. All results are highly significant and stable across specifications.

5.5. Network sensitivity analyses

The peer effects above are large and precisely estimated. This subsection checks how sensitive they are to the way I define the peer networks and second-degree peer instruments, applying five checks: 1) using random peers instead of the named ones, 2) dropping the most influential peers,

Table 6: Peer Institutions and the Probability of New Program Introduction

	N peers with program			Peer degree awards (100s)		
	(1) OLS	(2) OLS	(3) 2SLS	(4) OLS	(5) OLS	(6) 2SLS
Pr(Offered program in t | Not offered in $t-1$)						
1-year lag	0.001**	0.003**	0.006**	0.0006**	0.002**	0.004**
	(0.0001)	(0.0002)	(0.0004)	(0.0001)	(0.0003)	(0.0004)
Baseline mean	0.009	0.008	0.008	0.009	0.008	0.008
Effect as %	11.7%	40.6%	79.0%	7.0%	24.4%	46.5%
Major×Year FEs	✓	✓	✓	✓	✓	✓
School×Major FEs		✓	✓		✓	✓
First-stage F			705.6			276.6
N Observations	1,914,198	1,910,611	1,900,283	1,914,198	1,910,611	1,900,283
N Schools	1,706	1,693	1,682	1,706	1,693	1,682

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. The outcome is an indicator equal to one in the first year a school awards BA degrees in a given major, conditional on not having offered it previously; the program then exits the risk set. Treatment is lagged one year in all columns. Columns 1–3 use the raw number of peer institutions offering the major; columns 4–6 use total degree awards among peers in that field scaled to hundreds. Within each measure, the first column includes major-by-year fixed effects only, the second adds school-by-major fixed effects, and the third re-estimates the fully-specified model by 2SLS, instrumenting the peer measure with the corresponding peers-of-peers measure (the average among the school's second-degree peers), as in Section 5; the reported F is the first-stage statistic. Effect as % reports the coefficient divided by the baseline mean, times 100.

3) dropping peers that could affect a school outside its peer list, 4) trimming peers-of-peers most likely to resemble true peers based on a predicted peer-selection score, and 5) testing whether the effect is concentrated in particular kinds of networks, which speaks to the instrument's exclusion restriction. The peer effect estimates remain robust under the first four; the fifth suggests the estimates are stable aside from one real gradient by peer-list popularity, discussed below.

A primary concern is that the named peer lists carry no more information than any random set of schools would. Because nearly every school diversified its majors over this period, almost any group of schools might predict a school's EMI. To check this, I draw 100 random peer groups for each school, build placebo peer and second-degree peer EMI measures from them, and re-run the main estimates from Table 5. The random peer lists produce estimates centered on zero, while the real estimate is three to five times larger than even the most extreme placebo (Appendix Figure A4). The named peers appear to carry information that a random set of schools does not.

A second concern is that only the most prominent peers within a network drive the effects, so the list as a whole is less informative. I test this by dropping the single most influential peer from each school's list, and separately by dropping the most influential five percent of schools across the whole sample, where influence is how often a school is named as a peer by others (ties broken by Barron's selectivity and undergraduate FTE). Neither changes the substantive findings, and the estimates are if anything slightly larger with influential peers removed (Appendix Table A8).

A third concern speaks to the instrument's validity. The instrument assumes that peers-of-peers affect a school only through its own peers. Because peer lists run in one direction, a second degree peer of one focal school can also name that focal school as its own peer; in that case the peer-of-peers is plausibly a direct competitor and the instrument might act on the focal school through that relationship rather than through its peers. To address this, I keep the school's own peer list as is, but drop from the instrument any second-degree peer that names the school in return, and re-estimate.[25] The effect is essentially unchanged and the first stage stays strong (Appendix Table A8), so these reciprocal links do not appear to be driving the result.

A related concern is that the reciprocity screen only catches contamination that is directly observable in the data. Some $k(i)$ schools could be direct peers while neither focal nor $k(i)$ school listed one another explicitly. As a more conservative check, I estimate a peer-selection model on realized peer choices, $p(i)$, using the same characteristics associated with peer selection (relative enrollment, graduation rates, instructional spending, selectivity, distance, and sector), and use it to drop from $k(i)$ the schools most likely to resemble true peers by predicted score. This trim is intentionally aggressive because it removes schools that merely look like an explicit peer. Even dropping the most similar half of $k(i)$ by this measure, the elasticity remains large and statistically significant (1.2 to 1.3, compared to 1.98 in the baseline; Appendix Table A8).

The fifth check asks whether the peer effect is concentrated in particular kinds of networks or schools, which also bears on the instrument's exclusion restriction. If the peer correlation reflected a shared response to a common shock rather than genuine peer influence, the effect should weaken as networks become less similar or more diffuse, since a common shock plausibly hits similar schools alike but has no reason to travel through an arbitrary or distant peer list. I test this by interacting the peer effect with terciles of eight school and network measures: graduation rates, instructional spending, enrollment size, selectivity standing relative to peers, how often a school is named as a peer by others, average distance to peers, reciprocity, and clustering.[26]

Seven of the eight measures show no meaningful gradient: the peer elasticity sits within a tight band around the pooled estimate of 1.98 regardless of a school's graduation rate, spending,

[25] Roughly 5,700 pairs of schools are dropped during this test, which is about 2.5% of the total.

[26] Reciprocity is the share of first-degree peers that also consider the focal school a peer. Clustering is the extent to which a school's peers are also peers with one another, again as a share.

enrollment size, selectivity gap, reciprocity, or clustering (Appendix Table A9). This includes the measure most directly tied to a common-shocks story: the effect does not fade for schools with more distant or more loosely clustered peer networks, which is where that account would predict the weakest effect.

The one measure with a real gradient is peer-list popularity, or how often a school is named as a peer by others. Schools rarely named as a peer respond more strongly to their peers' diversification (elasticity 1.98) than schools frequently named as a peer (1.66), a gap of -0.318 (SE 0.078, joint $F = 8.68$, $p < 0.001$), which survives a Bonferroni correction for the eight moderators tested (threshold $p < 0.00625$). One reading is that infrequently-named schools behave as followers responding to a competitive set that is not, in turn, responding much to them, while frequently-named schools are more often the ones being followed. This is a difference in the size of the response rather than a case where the effect disappears in a dissimilar network, so it does not speak against the exclusion restriction. Taken together, these patterns support a peer response that is general across most school and network types, and the lack of fade in distant or loosely-clustered networks weighs against a common-shocks explanation for the peer correlation.

5.6. Alternative explanations

One potential alternative to a peer effects explanation of major diversification is a demand-driven one. As the labor market has tended to reward a broader range of skills, students may have moved toward more diverse fields and the schools serving them expanded to match what students and employers wanted. The correlation across peer institutions would be an artifact of a shared response to common conditions rather than my preferred explanation of peer competition. The concern relates to three separate but related issues beginning with 1) the broad and steady rise in demand for college graduates over this time period, 2) localized shifts in demand for particular skills and occupations, and 3) changes to what students themselves want from a college. I attempt to address each in turn.

The first concern is aggregate. Changing demand for college graduates over this period might simply have pulled more students into more fields regardless of peer institution imitation. I find little evidence of this. The relationship between occupational employment and major diversification, whether measured across all employment or just the share held by bachelor's-degree holders, and whether estimated by OLS or instrumenting for peers' EMI (Appendix Table C1, columns 2–5) is, at best, weakly positive. Nor can I attribute diversification to the *dispersion* of occupational demand: an inverse-Herfindahl index of bachelor's-level employment across occupations enters negatively and significantly under OLS—the wrong sign for a demand account, since it implies more diffuse occupational demand goes with *less* major diversification. Along similar lines, effective unemployment, weighted by a school's base-year mix of states from which it draws incoming students, shows no relationship to the EMI: an elasticity near 0.03 that is imprecise throughout (column 1).[27]

The second demand channel is more local. Colleges might add fields in response to shifts in demand for particular skills in their own markets, as a recent literature emphasizes (e.g., Conzelmann et al., 2023; Weinstein, 2022; Acton, 2021). At the program level, where this channel would show up most directly, an occupational demand shift-share carries no independent relationship with adoption of new programs, even once peer behavior enters the model. The peer measure estimates remain large and significant under both OLS and 2SLS (Appendix Table C6); field-specific demand does not displace the peer adoption effect.

For student demand-driven explanations, I lean on existing work that suggests willingness to pay for an additional program is small—about $50 per year in Cook (2021). Where demand shifts, college supply also tends to respond weakly, with a 10 percent increase in demand raising the number of related courses by just 2 percent (Light, 2026). A shared peer group response to student demand would also likely fade as peer groups grow more dissimilar and diffuse; the lack of heterogeneity described above (Appendix Table A9) goes against this.[28]

No design fully rules out a common demand shock correlated with peer structure. But the size of the effect, its robustness to random and influential-peer placebos, its stability across geographically distant networks, its persistence after screening out reciprocal ties, and its survival under an aggressive propensity-based trim of $k(i)$, are jointly hard to reconcile with a demand story that isn't also a peer effect.

6. Consequences of major diversification

The preceding section establishes that competitive peer pressure drives major diversification across baccalaureate institutions. Whether this expansion generates cost consequences for the institutions that undertake it is unclear. Rising instructional spending per student has attracted consid-

[27] This holds when instrumenting for unemployment with its lagged value to address measurement error.

[28] Full results from these demand robustness tests, including joint specifications that enter the competing explanations together, are in Appendix Section C. Graduate education spillovers and declining state support are examined there as well; both show effects small relative to peer effects, and neither accounts for the estimated peer correlations.

erable attention, with explanations ranging from labor-market cost disease (Archibald & Feldman, 2010; Baumol, 2012) to the prestige-seeking behavior that drives institutions to invest in services and programs (Blair & Smetters, 2021; Bowen, 1980; Clotfelter, 1996). I find a consistent positive association between curriculum diversification and instructional spending within school over time. I find new programs are substantially more expensive per credit hour than a school's existing ones at introduction. This, rather than reallocation from existing to new programs is a more likely source of cost increases.

6.1. Major diversification and instructional costs

Table 7 presents within-school estimates of the association between log EMI and instructional costs from two data sources. Panel A uses IPEDS data for approximately 1,700 institutions with log instructional expenditures per FTE as the outcome. Panel B uses the TCS sample with log cost per student credit hour as the outcome.[29] The TCS estimates in Panel B draw on the 97-institution perennial sample described in Section 3, which skews toward larger, public, doctoral-granting institutions relative to the full IPEDS sample; magnitudes in Panel B should be read with that composition in mind. Each panel reports three specifications: school and year fixed effects with no additional controls, with own student body characteristics and peers' average EMI added, and with peer institution characteristics added on top. Peers' average EMI enters in Columns 2 and 3 of each panel to ensure the own EMI coefficient does not absorb variation from competitive exposure.[30]

Across both datasets, own EMI is positively associated with measures of instructional expenditures and is stable across specifications. In IPEDS, a 10 percent increase in own EMI is associated with roughly half a percent higher instructional expenditures per FTE, significant at the five percent level or better in all columns (Panel A). In the TCS sample, the same increase in own EMI is associated with approximately 1.5 percent higher cost per student credit hour, marginally significant throughout (Panel B). Peers' average EMI is positive in both panels and mildly significant in some specifications, consistent with the peer mechanism documented in Section 5; the own EMI coefficient is stable to its inclusion helping to rule out a shared cost trend across the competitive network.

Applying these coefficients to the observed within-school increases in major diversity imply

[29] See Appendix B for detail on sample construction and the derivation of cost measures in Panel B.

[30] Standard errors are clustered at the institution level throughout. The IPEDS sample is weighted by total degrees; the TCS sample is weighted by total credit hours.

Table 7: Major Diversification and Instructional Costs

	Panel A: IPEDS *Log(instruct. expend. per FTE)*			Panel B: TCS *Log(cost per credit hour)*		
	(1)	(2)	(3)	(4)	(5)	(6)
Log(EMI)	0.043*	0.045**	0.053**	0.175	0.161	0.156*
	(0.018)	(0.017)	(0.015)	(0.096)	(0.088)	(0.075)
Log(peers' EMI)	–	0.040	0.088	–	0.495*	0.236
		(0.055)	(0.053)		(0.198)	(0.203)
Own controls		✓	✓		✓	✓
Peer controls			✓			✓
School FEs	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓
Observations	42,812	42,725	42,724	1,593	1,496	1,496
Institutions	1,699	1,697	1,697	97	91	91

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. IPEDS sample restricted to institutions with peer lists ($N \approx 1{,}700$); TCS sample restricted to institutions reporting consecutively for 10+ years ($N = 97$). Own controls: student body composition (part-time share, graduate share, female share, Black/Hispanic share, foreign share). Peer controls: peer averages of own controls. IPEDS weighted by total degrees; TCS weighted by total credit hours.

that EMI growth accounted for over 7 percent of the total increase in log instructional expenditures per FTE between 1990 and 2019 among peer-list institutions. An analogous calculation in the TCS sample, using Panel B's coefficient, implies a 23 percent contribution of the EMI toward cost per credit hour increases between 1998 and 2019.[31] Estimates from TCS are less precise than the IPEDS estimates, though the characteristics of TCS' sample is consistent with the complier profile described in Section 5 where institutions whose peer EMI was most responsive to the instrument were more likely to be public and moderately selective. The TCS estimate may thus be reflective of a population closer to where the underlying peer response is concentrated. Taken at face value, the association between major diversity and instructional costs is positive and economically meaningful, holding across two different datasets and measures of instructional costs.

To shed light on a potential mechanism, I compare the cost of new programs at introduction directly against the introducing school's existing portfolio. For each of the 162 program introduction events in the TCS sample with available cost data, I compute a cost ratio defined as the average cost per credit hour for the introduced field across other TCS institutions already established in

[31] Using the fitted Table 7 regression, this calculation compares each school's actual predicted spending in the final sample year to a counterfactual holding its EMI fixed at the first observed value. The resulting gap, averaged across institutions using the same weighting as the regression (degrees or credit hours), is expressed as a share of the total observed change over the same estimation sample and weighting, first year to last.

that field, divided by the introducing school's mean cost per credit hour across all its existing programs in the same year.[32] A ratio above one means the new program type typically costs more per credit hour at institutions where it is already established than the introducing school currently spends across its existing programs.

The mean ratio at introduction is 1.66: the introduced field, at institutions where it is already established, costs 66 percent more than the introducing school's own existing programs on average. Figure 6 reports mean ratios by broad field. Engineering (2.85) and Physical Sciences (2.54) carry the highest ratios, reflecting substantial equipment and faculty specialization costs. Communications is the only field below parity (0.67), as programs in this area tend to be introduced by institutions with existing portfolios that already exceed the communications benchmark. Health sciences and Business, which together account for roughly 38 percent of introduction events, carry moderate ratios of 1.41 and 1.11 respectively; the overall average reflects this frequency distribution more than the field-level extremes. A school engaged in sustained diversification is therefore perpetually carrying programs whose per-credit-hour cost exceeds that of its established operations. This effect, compounding across many introductions over many years, offers a direct account of the positive association between EMI and instructional costs in Table 7.

Importantly, the cost premium documented above coincides with peer-driven competitive pressure. I show this by classifying each introduction event by the number of programs introduced by the school's peers in fields the focal school *does not* yet offer in the 5 years leading up to introduction. Specifically, I use an indicator for above-median peer program introductions.[33] Introductions under high peer pressure carry a cost ratio 0.36 to 0.53 higher than those under low pressure (Table 8) against a mean cost ratio of 1.62 in this sample[34], over half of the entire premium above parity. This holds after controlling for the size of the introducing school's existing portfolio, normalizing peer counts, using a three-year window, and a version using all program introductions rather than only fields the school lacked.[35]

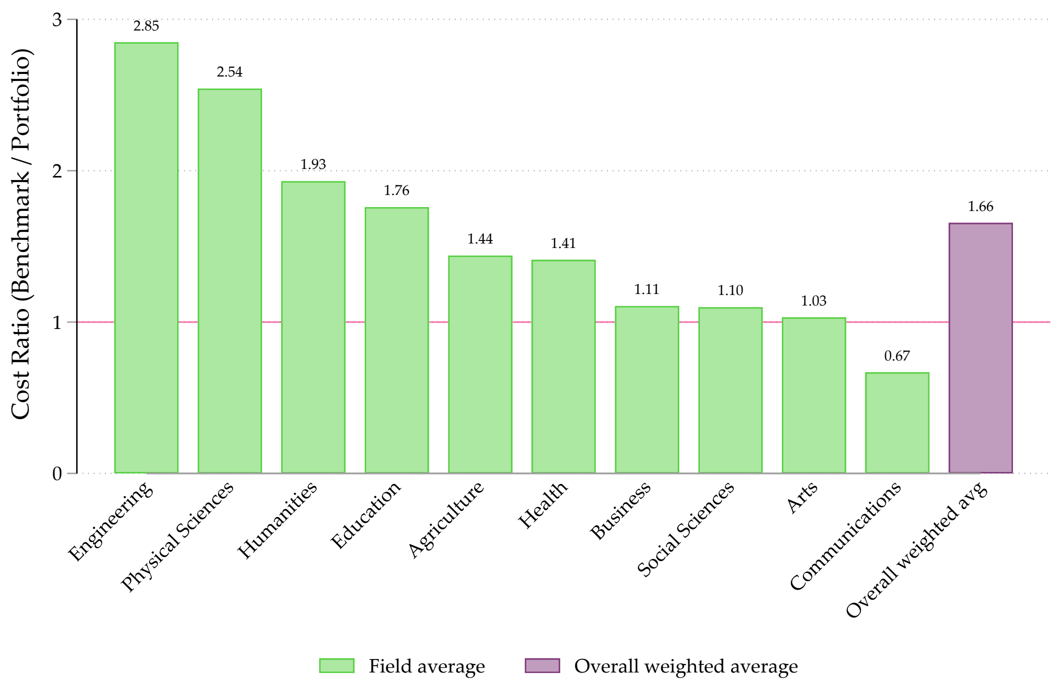
[32] The benchmark draws only on TCS institutions with at least three consecutive prior years of positive credit hours in the introduced field, ensuring it reflects established rather than early-phase costs at peer institutions. The 162 events are the subset of the 198 introduction events identified within the TCS analysis window for which cost data are available at the introduction year. Appendix Table D1 reports introduction counts and mean cost ratios by field.

[33] Construction detailed in Appendix D.

[34] This sample is slightly smaller than the original 162 due to fixed effects and removal of cells with single observations.

[35] Full robustness results can be found in Appendix Table D5 and Table D4.

Figure 6: Cost Ratio at Program Introduction, by Broad Field



Notes: Each bar shows the mean cost ratio at introduction for that broad field: the leave-one-out average cost per credit hour among TCS institutions already established in the introduced field, divided by the introducing school's own portfolio average cost per credit hour in the same year. The dashed line at one indicates parity. Ratios are weighted by the number of introduction events in each field. Sample restricted to the 162 of 198 identified introduction events with available cost data at the introduction year ($\tau = 0$).

Table 8: Peer Pressure and the Cost of Program Introduction

	1	2	3	4
High Peer Pressure	0.530*	0.361*	0.368*	0.400*
	(0.211)	(0.165)	(0.155)	(0.168)
Event-Year FE	✓	✓	✓	✓
Broad Field FE	✓			
Specific Major FE		✓	✓	✓
Selectivity (3-cat.)			✓	✓
Census Region				✓
Mean Cost Ratio (this sample)	1.619 (0.108)			
N Observations	146			
N Clusters	68			

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. Sample excludes introduction events at institutions that never submit a peer list to IPEDS, for whom peer pressure is not defined, and is held fixed across all four columns at the sample supported by the most restrictive specification (Specific Major FE). Outcome is the cost ratio at introduction (benchmark divided by portfolio, as in Figure 6). High Peer Pressure is a median-split indicator based on the number of programs, in any field, that a school's peers introduced in the five years before t that the focal school did not yet offer; construction detailed in Appendix D. Broad Field FE groups introductions into the same broad fields used in Figure 6 and Table D1. Specific Major FE uses the full 69-category major classification defined in Section 2 in place of broad field. Event-Year FE is the calendar year in which the introduction occurred. Selectivity (3-cat.) is the introducing institution's Barron's Competitiveness category. Census Region is the introducing institution's 4-category Census region. Mean Cost Ratio reports the unconditional sample mean of the outcome, with no covariates or fixed effects, for reference.

6.2. Program introductions and existing departments

The cost ratio evidence concerns new programs directly. A separate question is whether new programs also change costs or enrollment at the institution's pre-existing departments. I examine this using a stacked event study design (Cengiz et al., 2019) in the TCS panel, comparing log cost per credit hour, log credit hours, and log instructional expenditures at existing departments at introducing institutions against same-field departments at non-introducing TCS institutions over an eight-year window around each introduction event.[36]

Instructional expenditures and credit hours at existing bystander departments show no meaningful change after being exposed to a new program introduction: the pooled post-period averages are −0.010 (SE 0.012), 0.014 (SE 0.011), and 0.004 (SE 0.016) respectively, none statistically distinguishable from zero (Figure D1).[37] Pre-trend tests pass across all three outcomes. New program introductions do not definitively shift costs or instructional activity at departments already operating at the institution, at least four-years beyond introduction. The cost pattern documented in Table 7 does not operate by shifting enrollment away from existing departments or by changing their direct instructional costs (e.g., faculty salary or type).

The evidence in this section characterizes the cost implications of major diversification. Within-school associations in two independent datasets show that more diversified institutions spend more on instruction per student, and program-level cost comparisons identify a persistent cost premium associated with newly introduced fields as a plausible source.[38] Existing departments are unaffected on average. While the full causal structure of these relationships is not identified here, the results suggest a college that broadens its curriculum continuously over many years carries the cost of newly introduced programs whose per-credit-hour cost exceeds that of its established offerings, and that cost does not appear to be recovered through any efficiency adjustment in existing programs.

[36] See Appendix D for details of the event study design, including event definition, and the peer exclusion rule applied to the control pool.

[37] Joint F-tests across the five post-period coefficients are marginally significant for all three outcomes ($p = 0.034$, 0.036, 0.035; Appendix Table D2), reflecting oscillation across event years rather than a directional shift — no individual τ coefficient is itself significant, and there is no consistent sign pattern across the post-period window.

[38] As a further check, I test whether the introduction-year cost premium fades with program age. Point estimates are positive at every horizon across specifications; if anything, the direction is toward a rising one, though results are sensitive to specification choices and are not reported as supporting evidence.

7. Conclusion

Colleges and the market for baccalaureate education have evolved significantly over time. This paper documents a trend common to most baccalaureate-granting institutions in the United States: over the last several decades, the average college consistently expanded its major offerings, growing more diversified rather than specializing in a smaller number of core fields. The pattern reflects institutions responding to each other, driven by competition for students, resources, and prestige, more than to independent shifts in their own environments (Blair & Smetters, 2021; Clotfelter, 2017; Hoxby, 2009).

I illustrated the secular trend in major diversification across several data sources and showed it was predominantly driven by within-college changes to curricular offerings and prioritization of new programs. In tandem, the between-college variation in major diversity declined significantly, suggesting that smaller, less diverse colleges at the beginning of the panel (e.g., liberal arts) diversified at a faster rate, gaining significant ground on comprehensive research universities that traditionally offered the most diverse major options.

College behavior within this dimension was driven by peer effects. Consistent with other analyses of college decision-making (e.g., Acton et al., 2022; Blair & Smetters, 2021), they tended to care about their own relative positions and standing within groups of institutions with similar missions, resources, and students and responded to the average actions taken by those networks. This finding establishes a new way in which these peer effects manifest through the curricular offerings and priorities of colleges. This relates to other expositions of college mission drift and isomorphism (Jaquette, 2013; Morphew & Huisman, 2002). Typically construed as negative, I offer nuance to this notion: the diversification of degrees in accordance with peer institutions was perhaps necessary for smaller colleges to attract students and remain competitive with larger, more comprehensive institutions. This evidence pertains most directly to the roughly half of institutions that submit peer lists to IPEDS, which are larger, more selective, and predominantly non-profit, and which together award the large majority of bachelor's degrees in this period; whether similar peer dynamics operate among the smaller, less selective, and more often for-profit institutions that do not submit lists is not addressed here.

I showed major diversification is also associated with higher average instructional costs. This result adds something new to a persistent debate on the reasons for rising costs in higher education. In this setting, costs seem to rise because newly introduced programs enter in fields that cost roughly 1.66 times what the introducing school's existing portfolio costs on average, showing no evidence of fading with program age. This premium is larger specifically for programs introduced under high competitive pressure from peers, directly tying the cost consequences documented here to the peer mechanism established earlier in the paper. This is consistent with prestige and resource-dependency arguments for upward trends in costs. Future research could look more specifically at the characteristics of new program offerings and the contribution of different types of faculty labor (i.e., tenure vs. non-tenure track) to these cost premiums, and the degree to which institutions alter these choices when the college overall is diversifying.

The cost increases documented here should not be read as evidence that major diversification was welfare-reducing. A complete accounting would need to weigh these costs against benefits I do not estimate directly, including additional student choice, potentially higher rates of major completion, and better matches between students and fields of study. Existing micro-evidence on college course scarcity offers some guidance on how the availability of majors might mediate these outcomes (Kurlaender et al., 2014; Robles et al., 2021), and future work could also explore cross-disciplinary skill accumulation and its effects on long-run career outcomes, building on work like that of Han et al. (2023), particularly relevant given recent evidence on flattening lifetime earnings curves in STEM fields (Deming & Noray, 2020). I leave a full welfare accounting, potentially using a framework like the Marginal Value of Public Funds (Hendren & Sprung-Keyser, 2020), to future work.

Finally, this paper speaks to ongoing policy discussions and challenges within and across colleges in the United States. Many college leaders propose program elimination and downsizing when faced with financial difficulties (e.g., Quinn, 2020; Nietzel, 2026). While my analyses focus on the historical precedent of expanding programs, they still offer some insight into how the opposite might affect institutions and stakeholders in the aftermath of cost-cutting measures. If costs rise because newly introduced programs carry a persistent cost premium rather than a one-time start-up expense, program elimination could remove an ongoing cost driver. I find no evidence that new programs' premiums fade in early years of existence. Whether such measures actually lower costs, and what that means for students and the range of majors available to them, is an open question a full welfare accounting would need to address. This paper provides concrete evidence on how and why colleges expanded their major offerings, with room for further exploration on its precise consequences and extent to which colleges continue to diversify *or reverse* these trends in major supply.

References

Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of econometrics*, *113*(2), 231–263.

Acton, R. K. (2021). Community College Program Choices in the Wake of Local Job Losses. *Journal of Labor Economics*, *39*(4), 1129–1154.

Acton, R. K., Cook, E. E., & Luedtke, A. (2022). The influence of peer institutions on colleges' decisions: Evidence from fall 2020 reopening plans. *Journal of Economic Behavior & Organization*, *195*, 288–302.

Altonji, J. G., Arcidiacono, P., & Maurel, A. (2016). The Analysis of Field Choice in College and Graduate School: Determinants and Wage Effects. In E. A. Hanushek, S. Machin, & L. Woessmann (Eds.) *Handbook of the Economics of Education*, vol. 5, (pp. 305–396). Elsevier.

Andrews, R. J., Imberman, S. A., & Lovenheim, M. F. (2017). Risky Business? The Effect of Majoring in Business on Earnings and Educational Attainment. Working Paper 23575, National Bureau of Economic Research. Series: Working Paper Series.

Andrews, R. J., & Stange, K. M. (2019). Price Regulation, Price Discrimination, and Equality of Opportunity in Higher Education: Evidence from Texas. *American Economic Journal: Economic Policy*, *11*(4), 31–65.

Archibald, R. B., & Feldman, D. H. (2010). *Why Does College Cost So Much?*. Oxford University Press.

Autor, D., Dorn, D., Katz, L. F., Patterson, C., & Van Reenen, J. (2020). The Fall of the Labor Share and the Rise of Superstar Firms. *The Quarterly Journal of Economics*, *135*(2), 645–709.

Baumol, W. J. (2012). *The Cost Disease: Why Computers Get Cheaper and Health Care Doesn't*. Yale University Press.

Bianchi, N. (2020). The Indirect Effects of Educational Expansions: Evidence from a Large Enrollment Increase in University Majors. *Journal of Labor Economics*, *38*(3), 767–804. Publisher: The University of Chicago Press.

Blair, P. Q., & Smetters, K. (2021). Why Don't Elite Colleges Expand Supply? Working Paper 29309, National Bureau of Economic Research.

Bleemer, Z., & Mehta, A. (2021). College Major Restrictions and Student Stratification. Working Paper, UC Berkeley, Center for Studies in Higher Education.

Bleemer, Z., & Mehta, A. (2022). Will Studying Economics Make You Rich? A Regression Discontinuity Analysis of the Returns to College Major. *American Economic Journal: Applied Economics*, *14*(2), 1–22.

Blom, E., Cadena, B. C., & Keys, B. J. (2021). Investment over the Business Cycle: Insights from College Major Choice. *Journal of Labor Economics*, *39*(4), 1043–1082.

Bound, J., Lovenheim, M. F., & Turner, S. (2010). Why Have College Completion Rates Declined? An Analysis of Changing Student Preparation and Collegiate Resources. *American Economic Journal: Applied Economics*, *2*(3), 129–157.

Bowen, H. R. (1980). The Costs of Higher Education: How Much Do Colleges and Universities Spend per Student and How Much Should They Spend? Publisher: Jossey-Bass Inc.

Bramoullé, Y., Djebbari, H., & Fortin, B. (2009). Identification of peer effects through social networks. *Journal of Econometrics*, *150*(1), 41–55.

Brander, J. A., & Eaton, J. (1984). Product line rivalry. *The American Economic Review*, *74*(3), 323–334.

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230.

Cengiz, D., Dube, A., Lindner, A., & Zipperer, B. (2019). The Effect of Minimum Wages on Low-Wage Jobs. *The Quarterly Journal of Economics*, *134*(3), 1405–1454.

Chakrabarti, R., Gorton, N., & Lovenheim, M. F. (2020). State Investment in Higher Education: Effects on Human Capital Formation, Student Debt, and Long-term Financial Outcomes of Students. Working Paper 27885, National Bureau of Economic Research.

Cheslock, J. J., Ortagus, J. C., Umbricht, M. R., & Wymore, J. (2016). The Cost of Producing Higher Education: An Exploration of Theory, Evidence, and Institutional Policy. In M. B. Paulsen (Ed.) *Higher Education: Handbook of Theory and Research*, (pp. 349–392). Springer International Publishing.

Clotfelter, C. T. (1996). *Buying the Best: Cost Escalation in Elite Higher Education*. Princeton University Press. Publication Title: Buying the Best.

Clotfelter, C. T. (2017). *Unequal Colleges in the Age of Disparity*. Harvard University Press.

Conzelmann, J. G., Hemelt, S. W., Hershbein, B. J., Martin, S., Simon, A., & Stange, K. M. (2023). Skills, Majors, and Jobs: Does Higher Education Respond? Working Paper 31572, National Bureau of Economic Research.

Cook, E. E. (2021). Competing campuses: Equilibrium prices, admissions, and undergraduate programs in US Higher Education. Working paper.

d'Aspremont, C., Gabszewicz, J. J., & Thisse, J.-F. (1979). On hotelling's" stability in competition". *Econometrica: Journal of the Econometric Society*, (pp. 1145–1150).

de Chaisemartin, C., & D'Haultfœuille, X. (2020). Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review*, *110*(9), 2964–2996.

De Giorgi, G., Pellizzari, M., & Redaelli, S. (2010). Identification of Social Interactions through Partially Overlapping Peer Groups. *American Economic Journal: Applied Economics*, *2*(2), 241–275.

Deming, D. J., & Noray, K. (2020). Earnings Dynamics, Changing Job Skills, and STEM Careers. *The Quarterly Journal of Economics*, *135*(4), 1965–2005.

Denning, J. T., Eide, E. R., Mumford, K. J., Patterson, R. W., & Warnick, M. (2022). Why Have College Completion Rates Increased? *American Economic Journal: Applied Economics*, *14*(3), 1–29.

Ekerdt, L. K., & Wu, K.-J. (2023). The rise of specialized firms. Working Paper.

Gaynor, M., Ho, K., & Town, R. J. (2015). The industrial organization of health-care markets. *Journal of Economic Literature*, *53*(2), 235–284.

Gelbgiser, D., & Alon, S. (2016). Math-oriented fields of study and the race gap in graduation likelihoods at elite colleges. *Social Science Research*, *58*, 150–164.

Getz, M., & Siegfried, J. J. (1991). Costs and Productivity in American Colleges and Universities. In *Economic Challenges in Higher Education*, (pp. 259–392). University of Chicago Press.

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik Instruments: What, When, Why, and How. *American Economic Review*, *110*(8), 2586–2624.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277.

Han, S., LaViolette, J., Borkenhagen, C., McAllister, W., & Bearman, P. S. (2023). Interdisciplinary college curriculum and its labor market implications. *Proceedings of the National Academy of Sciences*, *120*(43), e2221915120.

Hearn, J. C., & Belasco, A. S. (2015). Commitment to the Core: A Longitudinal Analysis of Humanities Degree Production in Four-Year Colleges. *The Journal of Higher Education*, *86*(3), 387–416.

Hemelt, S. W., Hershbein, B., Martin, S., & Stange, K. M. (2023). College majors and skills: Evidence from the universe of online job ads. *Labour Economics*, *85*, 102429.

Hemelt, S. W., Stange, K. M., Furquim, F., Simon, A., & Sawyer, J. E. (2021). Why Is Math Cheaper than English? Understanding Cost Differences in Higher Education. *Journal of Labor Economics*, *39*(2), 397–435.

Hendren, N., & Sprung-Keyser, B. (2020). A unified welfare analysis of government policies. *The Quarterly journal of economics*, *135*(3), 1209–1318.

Hoxby, C. M. (2009). The Changing Selectivity of American Colleges. *Journal of Economic Perspectives*, *23*(4), 95–118.

Jaquette, O. (2013). Why Do Colleges Become Universities? Mission Drift and the Enrollment Economy. *Research in Higher Education*, *54*(5), 514–544.

Kessler, D. P., & McClellan, M. B. (2000). Is hospital competition socially wasteful? *The Quarterly Journal of Economics*, *115*(2), 577–615.

Kurlaender, M., Jackson, J., Howell, J. S., & Grodsky, E. (2014). College course scarcity and time to degree. *Economics of Education Review*, *41*, 24–39.

Leary, M. T., & Roberts, M. R. (2014). Do peer firms affect corporate financial policy? *The Journal of Finance*, *69*(1), 139–178.

Lee, L. (2003). Best Spatial Two-Stage Least Squares Estimators for a Spatial Autoregressive Model with Autoregressive Disturbances. *Econometric Reviews*, *22*(4), 307–335.

Light, J. (2026). Student demand and the supply of college courses. Working paper, Stanford University.

Manski, C. F. (1993). Identification of Endogenous Social Effects: The Reflection Problem. *The Review of Economic Studies*, *60*(3), 531–542.

Marginson, S. (2006). Dynamics of National and Global Competition in Higher Education. *Higher Education*, *52*(1), 1–39.

Melitz, M. J., & Polanec, S. (2015). Dynamic Olley-Pakes productivity decomposition with entry and exit. *The RAND Journal of Economics*, *46*(2), 362–375.

Millimet, D. L., & Bellemare, M. F. (2026). On the (Mis) Use of the Fixed Effects Estimator. *Oxford Bulletin of Economics and Statistics*, *88*(3), 484–505. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/obes.70031.

Moffit, R. A. (2001). Policy Interventions, Low-Level Equilibria, and Social Interactions. In S. N. Durlauf, & H. P. Young (Eds.) *Social Dynamics*. MIT Press.

Morphew, C. C., & Huisman, J. (2002). Using Institutional Theory to Reframe Research on Academic Drift. *Higher Education in Europe*, *27*(4), 491–506.

Newhouse, J. P. (1970). Toward a theory of nonprofit institutions: An economic model of a hospital. *The American economic review*, *60*(1), 64–74.

Nietzel, M. T. (2026). Academic purges continue as more colleges close their degree programs. *Forbes*. Accessed 7/7/2026.

Olley, G. S., & Pakes, A. (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, *64*(6), 1263.

Patnaik, A., Wiswall, M., & Zafar, B. (2021). College Majors. In B. P. McCall (Ed.) *The Routledge Handbook of the Economics of Education*. Routledge.

Quinn, R. (2020). West Virginia's Unprecedented Proposed Cuts Become Clear. *Inside Higher Ed*.

Robles, S., Gross, M., & Fairlie, R. W. (2021). The effect of course shutouts on community college students: Evidence from waitlist cutoffs. *Journal of Public Economics*, *199*, 104409.

Shaked, A., & Sutton, J. (1982). Relaxing price competition through product differentiation. *The Review of Economic Studies*, *49*(1), 3–13.

Sloane, C. M., Hurst, E. G., & Black, D. A. (2021). College Majors, Occupations, and the Gender Wage Gap. *Journal of Economic Perspectives*, *35*(4), 223–248.

Stange, K. (2015). Differential Pricing in Undergraduate Education: Effects on Degree Production by Field. *Journal of Policy Analysis and Management*, *34*(1), 107–135.

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, *225*(2), 175–199.

Teixeira, P., Rocha, V., Biscaia, R., & Cardoso, M. F. (2013). Competition and diversification in public and private higher education. *Applied Economics*, *45*(35), 4949–4958.

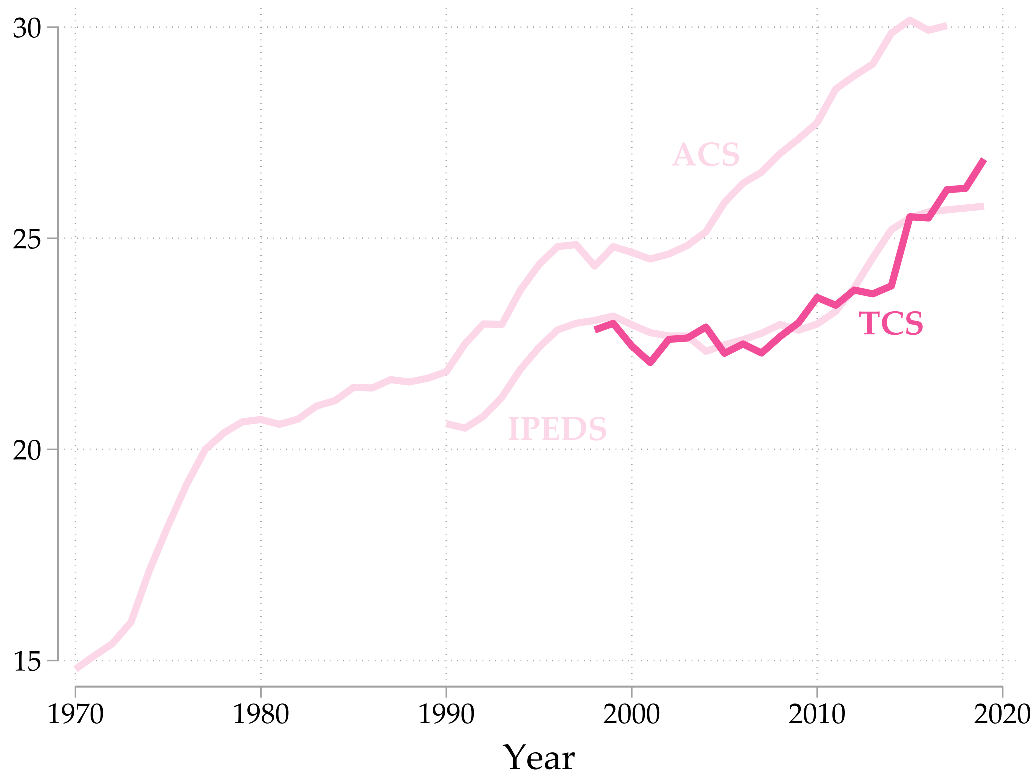
Thomas, J. (2024). What Do Course Offerings Imply about University Preferences? *Journal of Labor Economics*, *42*(1), 53–83.

Weinstein, R. (2022). Local Labor Markets and Human Capital Investments. *Journal of Human Resources*, *57*(5), 1498–1525. Publisher: University of Wisconsin Press Section: Articles.

Weisbrod, B. A., Ballou, J. P., & Asch, E. D. (2008). *Mission and Money: Understanding the University*. Cambridge: Cambridge University Press.

A. Additional Tables and Figures

Figure A1: The Cost Study EMI using student credit hours, 1998-2019



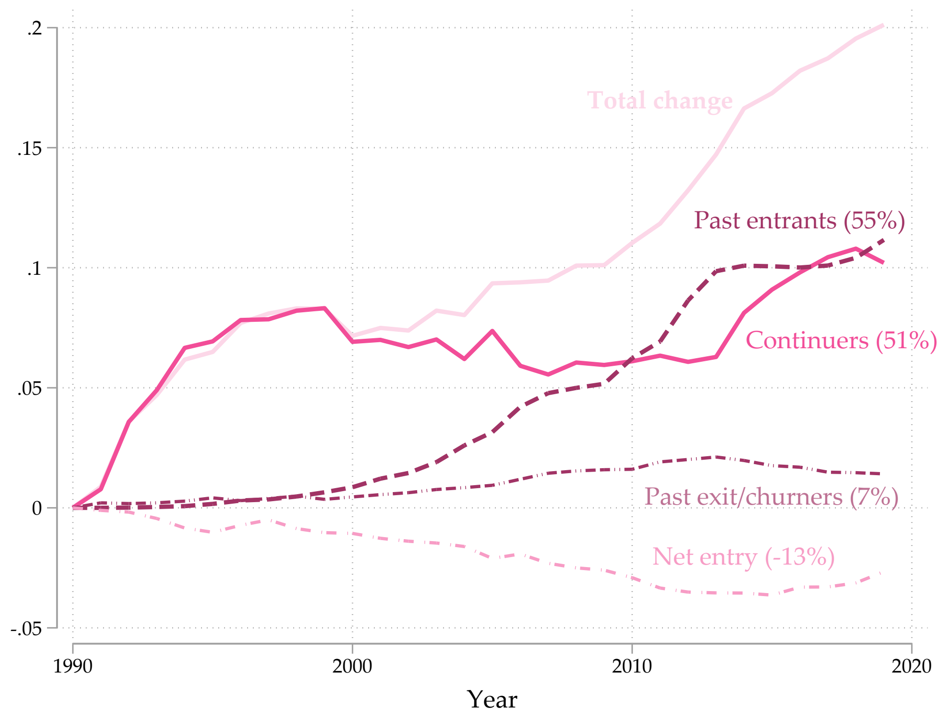
Notes: The EMI for the TCS sample is calculated using the squared shares of student credit hours taught in the 69 possible fields and summed across fields as opposed to bachelor's degrees as in IPEDS and the ACS. The original IPEDS and ACS EMI time-series are shown in lighter color for reference.

Table A1: Descriptive statistics for all fields in the analytic sample, 1990–2019

	Completion Share (%)			Offering Prevalence (%)				
Field	1990	2019	Δ	1990	2019	Δ	New	Exits
Accounting	4.3	2.5	-1.9	51.9	46.5	-5.3	304	112
Aeronautical Engineering	0.3	0.2	-0.1	3.3	2.9	-0.4	18	8
Agriculture	1.0	1.0	0.0	8.7	8.7	0.0	72	20
Allied Health	0.1	0.5	0.4	4.6	21.3	16.7	437	20
Applied Arts	2.1	2.1	0.0	48.3	50.2	1.9	390	51
Architecture	0.5	0.2	-0.3	5.1	4.5	-0.6	41	23
Atmospheric Sci & Meteorology	0.0	0.0	0.0	2.0	2.9	0.8	37	5
Biochem & Molecular Biology	0.2	0.6	0.4	9.3	25.9	16.7	464	19
Biology	3.2	5.2	2.0	65.4	59.2	-6.2	290	37
Biomedical Engineering	0.1	0.4	0.3	1.7	6.9	5.2	134	2
Business	12.6	10.8	-1.8	73.4	74.2	0.8	566	100
Chemical Engineering	0.3	0.5	0.2	8.6	7.2	-1.5	23	5
Chemistry	0.8	0.7	-0.1	50.9	45.9	-4.9	252	57
Civil Engineering	0.7	0.7	0.0	11.5	11.3	-0.2	75	10
Communication & Media Studies	3.3	3.4	0.1	40.4	50.0	9.6	531	61
Computer & Info Science	2.6	4.4	1.7	57.4	57.1	-0.3	470	127
Computer Engineering	0.2	0.5	0.3	4.6	12.2	7.5	214	10
Culinary Arts	0.0	0.0	0.0	0.0	1.5	1.5	34	0
Economics	2.3	1.9	-0.4	36.7	31.1	-5.6	179	87
Electrical Engineering	2.0	0.8	-1.2	15.4	14.8	-0.5	109	28
Engineering Technology	1.7	0.9	-0.8	16.2	14.9	-1.3	159	91
English, Liberal Arts, Humanities	9.4	5.4	-4.1	71.9	63.0	-8.9	302	72
Family & Consumer Sciences	1.3	1.2	-0.2	17.9	15.4	-2.4	160	108
Finance	2.6	2.2	-0.4	22.2	27.8	5.6	330	65
Fitness & Leisure Studies	0.4	2.6	2.1	13.9	38.1	24.2	679	30
Foreign Lang & Linguistics	1.1	1.1	0.0	41.5	38.7	-2.8	266	78
Geography	0.3	0.2	-0.1	14.6	11.3	-3.3	54	42
Geological & Earth Sciences	0.2	0.3	0.1	17.7	18.0	0.3	140	25
Health & Medical Admin Services	0.3	0.8	0.5	8.4	20.0	11.6	373	52
Hospitality Admin/Mgmt	0.6	0.5	0.0	8.4	9.0	0.6	128	62
Human Resources Mgmt & Services	0.4	0.5	0.1	7.9	13.2	5.2	233	63
Journalism	1.2	0.6	-0.6	17.5	14.2	-3.3	139	110
Legal Studies	0.2	0.2	0.0	5.6	9.8	4.2	180	48
Library Science	0.0	0.0	0.0	1.2	0.7	-0.5	14	18
Marketing	3.0	2.1	-0.9	25.4	31.2	5.8	371	82
Materials Science & Eng	0.0	0.1	0.0	2.3	2.3	0.1	26	11
Mathematics	1.3	1.3	-0.1	59.2	51.3	-7.9	246	69
Mechanical Engineering	1.4	1.8	0.4	13.3	14.7	1.4	125	12
Mental & Social Health Services	0.0	0.1	0.1	0.0	3.8	3.8	89	0
Mgmt Info Systems & Science	0.6	0.9	0.3	13.7	20.1	6.5	335	101
Microbiology	0.2	0.1	0.0	5.2	3.7	-1.4	22	24
Natural Resources	0.3	1.0	0.7	6.1	31.8	25.7	641	5
Nursing	2.7	7.1	4.4	32.4	45.0	12.6	552	61
Other Allied Health	0.9	1.9	1.0	28.7	29.3	0.6	328	139
Other Education	0.4	0.4	0.0	10.3	15.8	5.4	297	107
Other Engineering	1.1	1.0	-0.1	15.7	18.3	2.7	206	48
Other Majors	2.1	3.3	1.2	36.1	48.9	12.8	636	117
Other Physical Sciences	0.1	0.0	-0.1	10.6	2.4	-8.2	38	165
Other Social Sciences	1.8	1.8	0.0	43.7	36.3	-7.4	269	175
Other Visual/Performing Arts	1.7	2.3	0.7	52.0	52.6	0.6	398	67
Pharm Sciences & Admin	0.6	0.1	-0.5	3.8	1.8	-1.9	15	37
Philosophy & Religion	0.6	0.6	-0.1	41.1	38.2	-2.9	280	97
Physics	0.4	0.4	0.0	30.8	30.0	-0.8	215	45
Poli Sci/Gov & Intl Relations	3.7	2.4	-1.2	52.1	46.7	-5.4	240	49
PR & Advertising	0.4	0.8	0.3	6.1	14.9	8.8	271	28
Protective Services	1.5	2.8	1.3	22.5	39.9	17.4	605	62
Psychology	5.2	5.8	0.6	66.2	61.7	-4.5	345	47
Public Administration	0.3	0.2	-0.1	9.5	5.2	-4.2	49	90
Public Health	0.0	0.8	0.8	1.2	14.7	13.5	328	5
Public Policy	0.0	0.1	0.1	0.8	3.4	2.6	71	5
Rehab & Therapeutic Professions	0.6	0.2	-0.4	12.5	8.5	-4.0	118	135
Social Work	1.0	1.5	0.4	28.8	30.6	1.9	313	94
Sociology	1.5	1.4	-0.1	50.2	41.6	-8.6	205	100
Special Educ & Teaching	0.6	0.4	-0.2	18.1	17.9	-0.2	226	121
Statistics	0.0	0.2	0.2	3.9	6.4	2.5	92	9
Systems Engineering	0.0	0.0	0.0	0.7	1.0	0.3	16	5
Teacher Education	8.8	3.4	-5.5	62.0	50.2	-11.8	280	177
Theology	0.4	0.5	0.1	16.8	20.1	3.4	279	98
Urban Planning	0.4	0.2	-0.2	9.0	6.3	-2.6	56	63
Total	100.0	100.0	0.0	–†	–†	–†	16,380	4,096

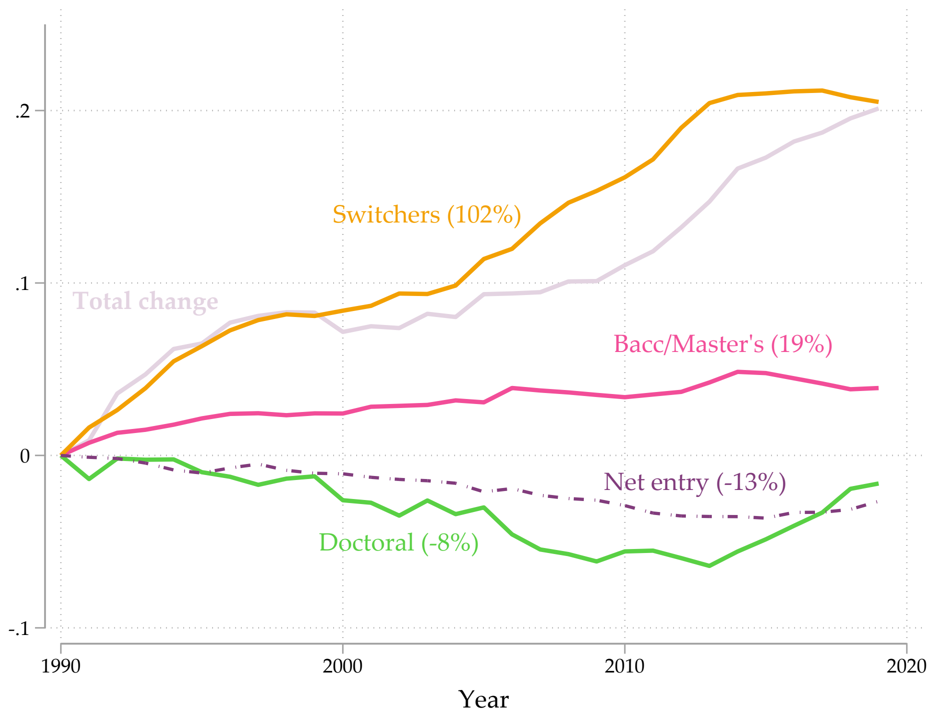
Notes: Each row corresponds to one of the 69 fields tracked in the analytic sample of four-year colleges. *Completion Share* is the field's share of total bachelor's degrees awarded at sample institutions; *Offering Prevalence* is the share of sample institutions awarding at least one bachelor's degree in that field. Both measures are reported as percentages ($\times$100). Δ denotes the change from 1990 to 2019 in percentage points. *New* is the count of new program introductions (field first offered at an institution) and *Exits* is the count of program discontinuations across the sample period. Fields are sorted alphabetically. New and Exits are counted at the school-by-field level, comparing 1990 and 2019 only. A "new" offering is a school-field pair with zero completions in 1990 and at least one in 2019; an "exit" is a school-field pair with at least one completion in 1990 and zero in 2019. Fields dropped and later re-added within the panel are not captured by this comparison. Completion shares sum to 100 percent by construction in both years, confirming the table covers the full sample. †Not reported; a simple average across fields would not reflect actual portfolio changes at the institution level. "Other Majors" comprises CIP 4-digit codes not classified into any named field, including military science and ROTC programs (CIP 28–29), vocational and trades programs (CIP 46–49), and small niche interdisciplinary specializations such as Medieval and Renaissance Studies and Peace Studies.

Figure A2: Decomposition of the change in log EMI, by fixed entry or exit status 1990-2019



Notes: This figure decomposes the total change in the log EMI similar to that depicted in Figure 4, only breaks continuing institutions into groups based on when they entered or exited the market for bachelor's degree awards.

Figure A3: Decomposition of the change in log EMI 1990-2019, by institution type



Notes: The cumulative contribution of each group is derived from the following yearly equation: $\Delta_1^2 Y = \text{net entry} + \sum_h \left(\frac{x_2^h}{x_2^C} \cdot Y_2^h\right) - \left(\frac{x_1^h}{x_1^C} \cdot Y_1^h\right)$, where Y_t^h is the (degree weighted) average log EMI for each continuing group, $h \in \{MB, D, S\}$, combining baccalaureate and master's institutions (MB). $D =$ Doctoral and $S =$ Switchers. Each point on the graph represents the cumulative sum of changes calculated for each group h. Changes are weighted by the group's overall share of continuing institution degrees in the respective time period. "Switchers" are those whose highest degree offering changed at some point between 1990 and 2019.

Table A2: Yearly Peer List Overlap

	Mean	SD	Institutions	min	max
Number of years peers submitted	8.735	2.140	1,719	2	10
Overlap Statistics					
2010 to 2011	0.960	0.155	1,384	0	1
2011 to 2012	0.970	0.138	1,515	0	1
2012 to 2013	0.983	0.101	1,523	0	1
2013 to 2014	0.950	0.182	1,506	0	1
2014 to 2015	0.961	0.171	1,541	0	1
2015 to 2016	0.949	0.179	1,457	0	1
2016 to 2017	0.976	0.124	1,480	0	1
2017 to 2018	0.975	0.118	1,445	0.0714	1
2018 to 2019	0.992	0.0638	1,438	0.0833	1
2011 to 2019	0.796	0.321	1,247	0	1
2010 to 2019	0.771	0.330	1,137	0	1

Notes: These estimates reflect the sample mean across institutions of a within-institution similarity of peer lists in year t and $t-1$. Called a Jaccard Index (J), this ranges from 0 to 1, and can be thought of in percentage terms. A J of 0.9, for example, would indicate a 90 percent overlap between lists in year t and $t-1$. Institutions that selected peers are those who submitted lists of peer institutions as part of IPEDS reporting and Data Feedback Reports. I include only schools who submitted these lists 2 or more times between 2010 and 2019.

Table A3: First Stage Results from IV Peer Estimates

	Year FE		State×Year FE		Linear Trend	
	1	2	3	4	5	6
	$\overline{Y}_k$	$\overline{\mathbf{X}}_k$	$\overline{Y}_k$	$\overline{\mathbf{X}}_k$	$\overline{Y}_k$	$\overline{\mathbf{X}}_k$
First Stage Coefficients, Treatment = Log($\overline{Y}_p$)						
Peers of peers' log(EMI)	0.586**		0.547**		0.687**	
	(0.056)		(0.054)		(0.049)	
Share FTE part-time		0.514**		0.574**		0.621**
		(0.166)		(0.161)		(0.156)
Share FTE graduates		0.541**		0.391*		0.634**
		(0.163)		(0.167)		(0.158)
Share BA's female		-0.165		-0.274		-0.934**
		(0.258)		(0.249)		(0.191)
Share BA's Black or Hispanic		0.349		0.437*		0.738**
		(0.190)		(0.177)		(0.125)
Share BA's Foreign		-1.489**		-1.421**		-0.905**
		(0.169)		(0.171)		(0.135)
Log average UG FTE		-0.134		-0.134		-0.172**
		(0.074)		(0.068)		(0.063)
Institutional age		0.032		0.038		-0.022
		(0.036)		(0.032)		(0.023)
Institutional age squared		-0.002*		-0.002**		-0.001
		(0.001)		(0.001)		(0.000)
First-stage F	110.0	22.3	101.9	23.7	194.6	20.5
N Observations	42,903	42,903	42,877	42,877	42,903	42,903
N Clusters	1,690	1,690	1,690	1,690	1,690	1,690

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All specifications include institution fixed effects. All models are weighted by the base-year total number of BA degrees awarded. Columns 1, 3, and 5 present first-stage estimates where the instrument for average peers' log EMI is the peers of peers' average log EMI, $\overline{Y}_k$. Columns 2, 4, and 6 present first-stage estimates where the instruments are represented by $\overline{\mathbf{X}}_k$ in the main text and include the average FTE undergraduate enrollment, average share of FTE part-time, graduate students, the share of BA's awarded to female students, Black or Hispanic students, and foreign students, and the average institutional age and its square. All models also include controls for the institution's own and peers' average log undergraduate enrollment, share of a school's total enrollment that is part-time, share graduate students, the share of bachelor's degrees awarded to women, to students that were either Black or Hispanic, to foreign or international students.

Table A4: Peer Effects Robustness: Fixed Effects vs. 6-Year Long Differences

	Panel A. OLS		**Panel B. IV**			
	(1)	(2)	(3)	(4)	(5)	(6)
	FE	6-yr LD	FE	FE	6-yr LD	6-yr LD
			2SLS	GMM 2S	2SLS	GMM 2S
Peers' Log(EMI) or $\Delta_{t_0}^{t_6}$Log(EMI)	0.311**	0.217**	1.978**	1.207**	2.382**	1.553**
	(0.082)	(0.051)	(0.291)	(0.269)	(0.458)	(0.302)
Instrument(s)			$\overline{Y}_k$	$\boldsymbol{\overline{X}_k}$	$\overline{Y}_k$	$\boldsymbol{\overline{X}_k}$
First-stage F			110.0	22.3	41.5	27.4
N Observations	43,057	32,880	42,903	42,903	32,880	32,880
N Clusters	1,702	1,602	1,690	1,690	1,602	1,602

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All models are weighted by the base-year total number of BA degrees awarded. Panel A presents OLS estimates. Panel B presents IV estimates. Columns 3 and 5 instrument for peers' log(EMI) or 6-year change in log(EMI) using the analogous average outcome of excluded second-degree peers ($\overline{Y}_k$); columns 4 and 6 use average covariate values of excluded second-degree peers ($\boldsymbol{\overline{X}_k}$), estimated via GMM 2S following Lee (2003). Fixed effects specifications (columns 1, 3, 4) include school and year fixed effects. Long difference specifications come from the following general estimating equation: $\Delta_{t_0}^{t_6} Y_i = \gamma + \beta_2 \Delta_{t_0}^{t_6} \overline{Y}_{p(i)} + \boldsymbol{\Gamma}\left(\boldsymbol{\Delta_{t_0}^{t_6} X_i}\right) + \boldsymbol{\Pi}\left(\boldsymbol{\Delta_{t_0}^{t_6} \overline{X}_{p(i)}}\right) + \eta_{t_0} + \Delta_{t_0}^{t_6}\epsilon_i$. (columns 2, 5, 6), and include base-year fixed effects; 6-year differencing removes school-level fixed characteristics over each interval. All specifications include the full control vector described in Table 5 notes. These long differences attempt to address potential concerns that fixed effects estimates may be biased in long panels (here T=30) when unobserved institutional heterogeneity is persistent but not strictly time-invariant.

Table A5: Institutional Characteristics of Peer-Effect Compliers

	Complier Mean	Full-Sample Mean	Difference
Selectivity (Barron's)			
High	0.172	0.197	-0.025
Moderate	0.731	0.641	0.090
Low	0.097	0.162	-0.065
Institutional control			
Public	0.781	0.647	0.134
Private nonprofit	0.195	0.320	-0.125
For-profit	0.024	0.033	-0.009
Highest degree offered			
Bachelor's (no graduate programs)	0.097	0.065	0.032
Master's	0.279	0.253	0.026
Doctoral	0.624	0.682	-0.059
Size, resources, and outcomes			
Undergraduate enrollment (FTE)	13,900	14,200	-300
Instructional expenditure per FTE ($)	11,100	12,000	-900
Average distance to peers (miles)	731	716	16
Peer-list popularity (times named as a peer)	27.5	26.8	0.7
150%-time graduation rate	0.603	0.588	0.016
Estimated complier share of sample, $E[\kappa]$	0.216		
N		42,903	

Notes: Complier characteristics are estimated using the kappa-weighting method described in Section 5 (see main text footnote for full citation). The underlying specification relies on a binary treatment and instrument, where peers' EMI ($\overline{Y}_p$) and peers-of-peers' EMI ($\overline{Y}_k$) was split at its median. Both the propensity score and the reported averages are computed using total degrees as weights, matching the main regression specifications. The complier mean for each characteristic is the kappa-weighted average across the estimation sample; the full-sample mean is the degree-weighted average over the same sample for comparison. $E[\kappa]$ reports the estimated complier share of the sample. Instructional expenditure and graduation rate have smaller N (42,486 and 29,469 respectively) due to missingness in those variables specifically; all other rows use the full N=42,903.

Table A6: Robustness of Peer Effect Estimates to Specification Choices

	Peers' Log(Avg. EMI)	SE	First-stage F	Hansen's J	(p)	N
Panel A: $\overline{Y}_k$ ***instrument (2SLS)***						
Preferred spec (year FE)	1.978**	(0.291)	110.0			42,903
State-specific linear trend	1.676**	(0.214)	180.1			42,903
Lag 1 (year FE)	1.993**	(0.298)	98.0			41,133
Lag 2 (year FE)	1.964**	(0.299)	89.0			39,461
Unweighted (year FE)	1.338**	(0.315)	28.5			42,903
Selectivity tier × year	2.026**	(0.296)	115.5			42,903
Expenditure decile × year	2.051**	(0.363)	73.8			42,492
Graduation-rate decile × year[a]	1.986**	(0.248)	175.9			29,444
Panel B: $\overline{X}_k$ ***instrument (GMM 2S)***						
Preferred spec (year FE)	1.207**	(0.269)	22.3	6.895	0.440	42,903
State-specific linear trend	1.410**	(0.247)	23.2	6.295	0.506	42,903
Lag 1 (year FE)	1.362**	(0.308)	20.6	6.717	0.459	41,133
Lag 2 (year FE)	1.385**	(0.351)	17.4	6.921	0.437	39,461
Unweighted (year FE)	0.476**	(0.182)	7.5	13.900	0.053	42,903
Selectivity tier × year	1.247**	(0.286)	20.8	6.374	0.497	42,903
Expenditure decile × year	0.879**	(0.297)	14.2	6.244	0.512	42,492
Graduation-rate decile × year[a]	0.998**	(0.253)	17.3	11.225	0.129	29,444

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses (N clusters omitted here; see Table 5 for the corresponding cluster counts). Each row varies a single dimension of the preferred specification in Table 5, column 2 (or 3 for Panel B), holding all else fixed. "State-specific linear trend" replaces year fixed effects with school-invariant state-by-year linear trends. "Lag" refers to the number of years by which the peer EMI measure is lagged relative to the outcome. "Unweighted" drops the base-year total BA degree weight used elsewhere. "Selectivity tier," "expenditure decile," and "graduation-rate decile" replace year fixed effects with fixed effects for, respectively, Barron's selectivity category, decile of instructional expenditures per FTE, and decile of six-year graduation rate, each interacted with year.

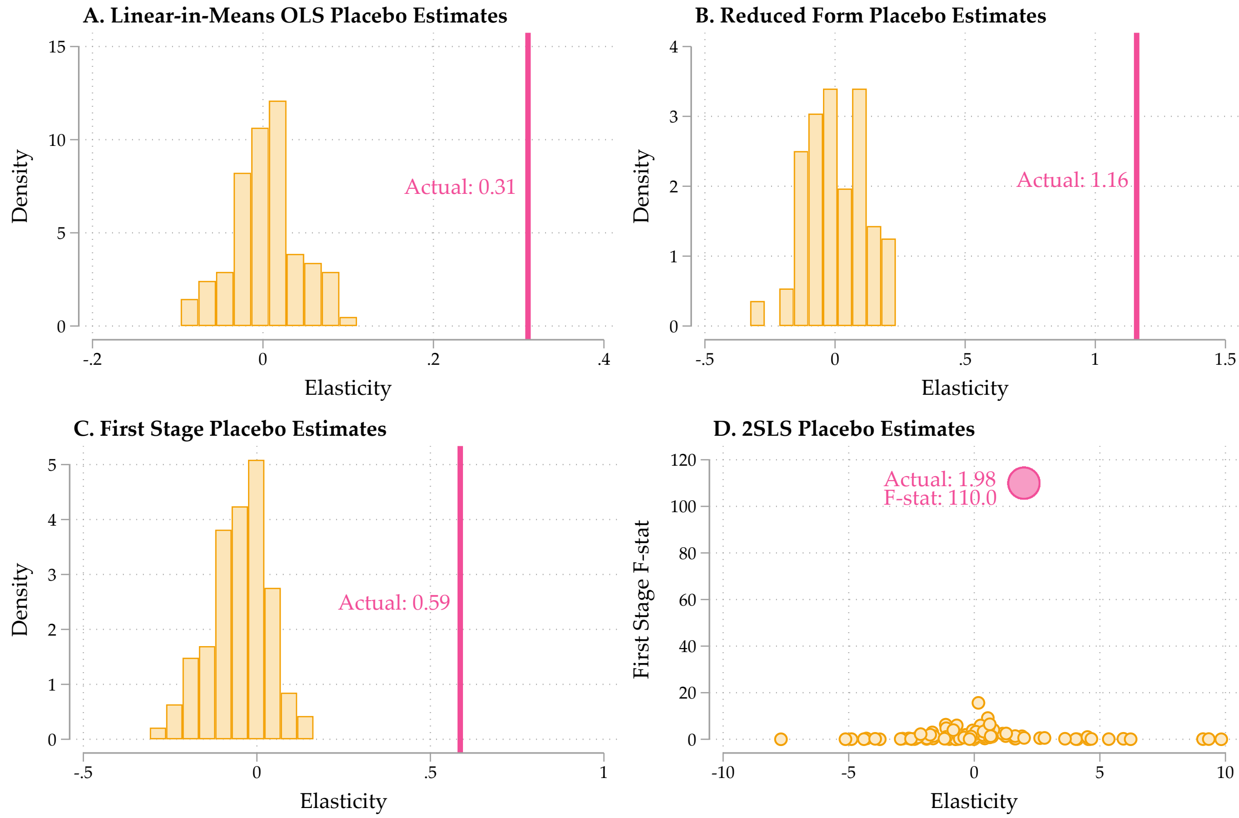
[a] Graduation rate is unavailable for a subset of the panel, reducing the estimation sample relative to the other rows.

Table A7: Peer Effect Estimates Restricted to the Observed Network Window (2010–2019)

	Peers' Log(Avg. EMI)	SE	First-stage F	Hansen's J	(p)	N
Panel A: $\overline{Y}_k$ ***instrument (2SLS)***						
Year FE	2.079	(1.531)	5.3			15,752
Linear year trend	1.509*	(0.656)	27.5			15,752
State-specific linear trend	1.584**	(0.554)	43.8			15,752
State$\times$year FE	1.950	(1.266)	8.1			15,742
Panel B: $\overline{X}_k$ ***instrument (GMM 2S)***						
Year FE	0.415	(0.252)	13.7	10.643	0.155	15,752
Linear year trend	0.612*	(0.262)	14.0	9.880	0.195	15,752
State-specific linear trend	0.671*	(0.278)	14.6	10.507	0.162	15,752
State$\times$year FE	0.414	(0.266)	15.9	10.439	0.165	15,742

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses (1,681 clusters in all rows except state$\times$year FE, which has 1,680). All models weighted by the base-year total number of BA degrees awarded, lag 0. This table re-estimates the specification in Table 5 restricted to 2010–2019.

Figure A4: Placebo Peer Estimate Distribution with Actual Peer Group Estimates



Notes: Each panel depicts estimates generated using 100 randomly drawn peer groups and the resulting excluded peers of peers, with the focal institution's actual log EMI as the dependent variable throughout. The vertical line in each panel marks the estimate obtained from the actual peer lists. Panel A plots the elasticities from re-estimating the linear-in-means OLS model of Equation 5, reported in column 1 of Table 5. Panel B plots the reduced-form estimates, regressing own log EMI on the placebo peers-of-peers average log EMI ($\overline{Y}_{k(i)}$). Panel C plots the corresponding first-stage estimates, regressing the placebo peers' average log EMI on the placebo peers-of-peers average log EMI. Panel D plots the 2SLS estimates instrumenting the placebo peers' average log EMI with $\overline{Y}_{k(i)}$, as in column 2 of Table 5; the actual estimate is 1.98 with a first-stage F of 110.0.

Table A8: Peer Effect Under Alternative Peer-Group and Instrument Definitions

	OLS	2SLS ($\overline{Y}_k$)
Main (full peer list)	0.311**	1.978**
	(0.082)	(0.291)
First-stage F		110.0
Changes to main peers, $p(i)$		
Drop most-influential peer	0.278**	1.995**
	(0.077)	(0.275)
First-stage F		124.4
Drop top-5% influential peers	0.253**	3.884**
	(0.069)	(1.132)
First-stage F		12.2
Changes to peers of peers, $k(i)$		
Drop second-degree peers listing focal inst.	—	2.000**
		(0.308)
First-stage F		97.0
Trim top 10% by predicted peer-selection score	—	1.200**
		(0.207)
First-stage F		169.4
Trim top 50% by predicted peer-selection score	—	1.312**
		(0.318)
First-stage F		51.4

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. Each row re-estimates the effect of peers' average log(EMI) on own log(EMI) under a different definition of the peer group or instrument, holding the column estimator fixed. All specifications include institution and year fixed effects, are weighted by the base-year total number of BA degrees awarded, use the control vector from Table 5, and are restricted to institutions with peer lists. The 2SLS column instruments peers' log(EMI) with the average log(EMI) of excluded second-degree peers ($\overline{Y}_k$); the reported F is the first-stage statistic on that instrument.

"Main" uses the full self-reported peer list. "Drop most-influential peer" removes the single most frequently named peer from each school's first- and second-degree peer sets (ties broken by Barron's selectivity and undergraduate FTE). "Drop top-5% influential peers" removes the five percent of institutions most frequently named as peers system-wide from both sets. "Drop second-degree peers" leaves the first-degree peer set unchanged but drops from the instrument any second-degree peer that itself names the focal institution as a peer.

Rows under *Changes to peers of peers*, $k(i)$ hold the main peer list $p(i)$ fixed and instead modify the excluded second-degree peer set used to construct $\overline{Y}_k$; OLS is not defined for these rows since $k(i)$ does not enter the OLS specification. The peer-selection score is the predicted probability from a logit model of realized peer choice, estimated on focal-candidate characteristic gaps in enrollment, graduation rate, instructional expenditure per FTE, selectivity, distance, and sector.

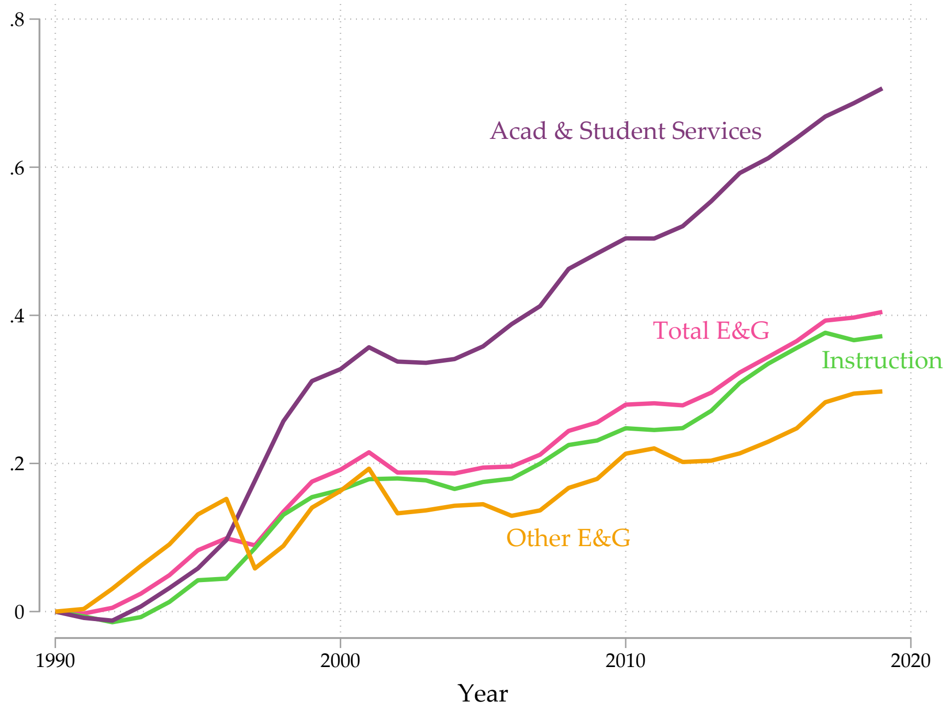
Table A9: Peer Effect Heterogeneity Across Network and Outcome Moderators

	Peer effect, by tercile			Heterogeneity tests	
	T1 (low)	T2 (mid)	T3 (high)	T3−T1 gap	Joint F (p)
Graduation rate ratio (own/peer average)					
Peers' Log(Avg. EMI)	1.934**	1.933**	1.934**	−0.000	0.23
(SE; p)	(0.239)	(0.239)	(0.239)	(0.005)	(0.794)
First-stage F	183.1	183.5	183.3		
Instructional expenditure ratio (own/peer average)					
Peers' Log(Avg. EMI)	1.938**	1.945**	1.952**	0.014	4.30
(SE; p)	(0.288)	(0.288)	(0.289)	(0.005)	(0.014)
First-stage F	113.2	112.8	112.2		
FTE enrollment ratio (own/peer average)					
Peers' Log(Avg. EMI)	1.989**	1.983**	1.976**	−0.013	0.90
(SE; p)	(0.292)	(0.292)	(0.293)	(0.012)	(0.407)
First-stage F	114.4	113.6	112.5		
Peer-list popularity (times listed as a peer by others)					
Peers' Log(Avg. EMI)	1.978**	1.857**	1.660**	−0.318	8.68
(SE; p)	(0.271)	(0.273)	(0.277)	(0.078)	(<0.001)
First-stage F	120.6	119.2	120.2		
Average peer distance (miles)					
Peers' Log(Avg. EMI)	2.040**	1.924**	1.915**	−0.125	1.11
(SE; p)	(0.286)	(0.302)	(0.297)	(0.084)	(0.331)
First-stage F	122.3	113.9	113.3		
Selectivity standing (signed gap vs. peers)					
Peers' Log(Avg. EMI)	1.871**	1.972**	1.730**	−0.142	2.32
(SE; p)	(0.280)	(0.265)	(0.251)	(0.128)	(0.099)
First-stage F	144.1	174.2	136.8		
Reciprocity (share of peers that list focal in return)					
Peers' Log(Avg. EMI)	2.165**	1.960**	1.906**	−0.259	1.48
(SE; p)	(0.323)	(0.308)	(0.285)	(0.151)	(0.229)
First-stage F	107.3	129.0	115.8		
Clustering (share of peer pairs that are themselves peers)					
Peers' Log(Avg. EMI)	1.999**	1.956**	1.993**	−0.006	0.06
(SE; p)	(0.311)	(0.285)	(0.344)	(0.144)	(0.946)
First-stage F	104.1	119.7	118.0		

Notes: * $p < 0.05$, ** $p < 0.01$. The table reports 2SLS estimates of the effect of peers' average log(EMI) on own log(EMI), allowing the peer effect to differ across terciles of each moderator listed in italics. All tercile slopes are estimated in a single regression with a common control vector and a common set of institution and year fixed effects. Each tercile's peer term is instrumented by the corresponding tercile interaction of the excluded peers-of-peers average log(EMI) ($\overline{Y}_k$). First-stage statistics are the Sanderson-Windmeijer multivariate F for each tercile-specific endogenous regressor. Standard errors clustered at the institution level in parentheses. All models follow the specification in Table 5, column 2, weighted by the base-year total number of BA degrees awarded and restricted to institutions with peer lists. "T3−T1" reports the difference in the high- and low-tercile peer slopes with its standard error and p-value, and "joint equality" reports the test that all three tercile slopes are equal.

The graduation rate, instructional expenditure, and FTE enrollment moderators are ratios of the focal institution's value to the average across its peers; values above one indicate the focal institution exceeds its peer average. Peer-list popularity counts the number of times an institution is named as a peer by others. Average peer distance is the mean geographic distance from the focal institution to the institutions on its peer list. Selectivity standing is the mean across peers of a signed indicator equal to $+1$ when the focal institution is more selective than a peer, -1 when less selective, and 0 at the same Barron's Competitiveness Index value, so that higher values indicate an institution standing above its peers in selectivity. Reciprocity is the share of an institution's listed peers that list it in return, and clustering is the share of peer pairs on an institution's list that also name one another.

Figure A5: Cumulative log change in Educational and General expenditures, by type 1990-2019



Notes: Source: IPEDS finance files, harmonized across years and institution types. All expenditures were adjusted to 2019 terms using the CPI prior to calculating changes. educational and general (E&G) excludes operation and maintenance of plant expenditures in this paper. The "Other" category includes research, institutional support, public service, and scholarship and fellowship expenditures

B. Data Appendix

B.1. IPEDS

As mentioned in the main text, the IPEDS sample is limited to all schools and years between 1990 and 2019 in which at least one bachelor's degree was awarded. This includes public, private non-profit, and for-profit colleges in all 50 states and Washington, D.C., but I exclude schools in Puerto Rico and other outlying islands and territories. These exclusions are made mainly because data on other constructs in the paper, like unemployment and state financial aid, are unavailable for these areas. I use institution identification number crosswalks from the Department of Education's College Scorecard dating from 2000 through 2019 to accommodate reporting changes within institutions' IPEDS identification numbers. This accounts for schools that merged with other schools either physically, or only for reporting purposes in IPEDS. I treat merged institutions as one entity so as not to inflate the contribution of net entry and exit in the changes to the EMI. Merging of institutions could be analyzed in more detail on its own to examine how major offerings and diversity changed after such events, though this is outside the scope of this particular paper.

The bachelor's degree completion data are reported at the 6-digit CIP-code level. Given the length of the panel, there are several vintages of codes that were used in reporting. The National Center for Education Statistics (NCES) typically introduces a new version of the codes each decade, with the exception of 1985. The 1985 version was also used by some institutions to report degrees in the 1990 and 1991 degree data from IPEDS. In order to make cross-decade comparisons of major diversity without artificially inflating majors due to the addition of new CIP codes over time, I harmonize CIP vintages to obtain a constant set of 4-digit CIP codes across all years of data. This typically involved a least-common-denominator approach to crosswalking codes that were added or changed: if one code was expanded to two separate codes, or vice versa, I use the less specific code across both years of data. With the harmonized list of 4-digit codes, I further collapse these to the 69 categories used in the main text. Hemelt et al. (2023) describes the aggregate codes in more detail. These categories are sufficiently detailed so as to preserve the CIP code organization structure, but broad enough to capture a large number of degrees while also eliminating issues of artificial field inflation, described in more detail in the next paragraph.

The choice of aggregation level and the treatment of second majors could each, in principle, affect measured diversity. Appendix Table B1 shows the aggregate EMI trend and the headline peer elasticity are stable to collapsing to 2- or 4-digit CIP codes instead of the 69-category scheme used throughout the paper. The table also shows stability to down-weighting or excluding second-major completions (IPEDS MAJORNUM=2) in the post-2001 subsample for which first- and second-major records are reliably distinguished; this restricted subsample is naturally less powered than the full panel, so the corresponding first-stage F is reported alongside the estimates.

Table B1: Sensitivity to Degree-Counting Choices

	Major Definition (t_0=1990)			Second Majors (t_0=2001)		
	Main (69-cat.)	2-digit CIP	4-digit CIP	*Main* (w=1)	w=0.5	w=0
$\Delta_{t_0}^{2019}$ log EMI	0.201	0.166	0.160	0.126	0.122	0.117
2SLS elasticity	1.978**	1.426**	2.234**	3.836*	3.490*	3.169**
	(0.291)	(0.173)	(0.415)	(1.687)	(1.396)	(1.158)
First-stage F	110.0	231.6	62.1	6.8	9.1	12.3
N Observations	42,903	42,903	42,903	28,848	28,848	28,848
N Clusters	1,690	1,690	1,690	1,688	1,688	1,688

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All models are weighted by the base-year total number of BA degrees awarded. The 2SLS specification matches the preferred column of Table 5: school and year fixed effects, instrument is average log(EMI) among excluded peers of peers ($\overline{Y}_k$). *Major Definition* columns vary how bachelor's degree completions are categorized when constructing EMI—using 69 custom categories (main specification), 2-digit CIP codes, or 4-digit CIP codes—holding the peer network and all controls fixed; the full 1990–2019 panel is used. *Second Majors* columns recompute EMI weighting IPEDS MAJORNUM=2 (second-major) completions by $w \in \{0, 0.5, 1\}$ and restrict the panel to 2001–2019, the first year IPEDS reliably distinguishes first from second majors; *Main* (w=1) matches the main specification used throughout the paper. $\Delta_{t_0}^{2019}$ log EMI is the difference in degree-weighted mean log(EMI) between year t_0 and 2019.

The harmonization of codes was most prominent in fields that went through significant change throughout the decades of interest. Computer Science, as one example, had just 6 4-digit CIP codes in 1985. By 2010, this nearly doubled to 11, reflecting the technological advances and increased number of occupations and tasks associated with this discipline. My harmonized version keeps the number at 6 codes, giving all new 4-digit codes in this area to the "Computer and Information Sciences, Other" catch-all code (11.99). The final major code collapses these all into 1, in many ways making the first step superfluous, but this may not always be the case. In general, this approach is conservative given the aims of this paper to quantify curricular diversification. It also bars against artificial inflation of diversity. Since new CIP codes are only released every decade, it is very likely colleges were awarding degrees in what eventually become "new" CIP codes, without any official designation. It is also very likely that colleges do not uniformly designate their degree awards across 4-digit CIP codes (let alone at the 6-digit level). For instance, some colleges transition to new codes slower than other colleges and the substantive differences across codes within the larger

2-digit codes are not usually denoted clearly by NCES. All these reasons should make it clear that examining major diversity at levels lower in code-specificity than I've done is difficult and potentially ill-advised, particularly when looking across a large group of institutions.

While some of the most central elements to this paper like degree completions, institution characteristics, and institutional expenditures and revenue are available for all 30 years, several other supplementary data elements are not available for all years and units of interest. I depict the years in which each data element was available in Figure B1. Some data elements from IPEDS were not collected until later in the panel, or were not collected on an annual basis. For example, six-year graduation rate cohorts start in 1991 (1997 collection year), omitting the 1990 base year of the panel. Information on the number of applicants, accepted students, and their test score information was first collected in 2002. The state of residence for incoming first-year students were collected every other year beginning in 1992.

FTE enrollment was not officially reported until 2004 onward. As FTE is a more precise estimate of enrollment for purposes of resource allocation I use fall enrollment counts of full- and part-time students in each year and institution to estimate FTE for all years in the panel. I assign weights of 1 to full-time students and 0.5 to part-time students and estimate the predictive power of this formula to capture the actual FTE estimates in the data when explicitly reported from 2004 to 2019. A linear regression of actual onto predicted FTE shows a coefficient of 0.97 and R-squared value of 0.96, suggesting this is a suitable prediction method. All references to FTE in the paper, regardless of year, use this formula-based FTE calculation to facilitate its use across all 30 years of interest.

B.2. The Cost Study at the University of Delaware

I use department-level data on instructional expenditures collected as part of the TCS in analyses of costs per credit hour after a new program introduction. These data are available for an unbalanced panel of institutions from 1998 through 2019. Hemelt et al. (2021) discusses the representation of these data in detail, though in general, participating institutions are more likely to be larger public research universities. In order to create a more balanced panel to analyze changes in average instructional costs within departments, I subset the full TCS dataset to institutions that participate for at least 10 consecutive years (out of 22). This yields 98 institutions from the original 622 in the full sample, though nearly half of all schools in the full sample participate just one or two years consecutively. One institution is excluded from all analyses due to data-quality concerns identified during diagnostic checks, leaving 97 institutions in the analytic sample.

In Table B2 I provide basic descriptive statistics of the full IPEDS sample, compared to the perennial and non-perennial TCS samples. The sub-analysis of cost spillovers in Section 6 is performed using these perennial TCS participants. In general, the perennial participants are overwhelmingly likely to be public and awarded from doctoral-granting institutions. In comparison, the full IPEDS sample of degrees awarded are 65 percent public and 74 percent doctoral granting across all years. The perennial participants also have higher average major diversity values than the full IPEDS sample. The samples are not markedly different in terms of demographic characteristics of their BA completers.

B.3. Employment data

I obtain occupational employment for each state and year from 1997 through 2018 from the Occupational Employment and Wage Statistics (OEWS) survey and unemployment rates from the Local Area Unemployment Statistics (LAUS) from 1990 through 2019, both from the Bureau of Labor Statistics (BLS). I harmonize the OEWS occupation codes across different vintages of Standard Occupational Classification (SOC) codes using resources from the National Crosswalk Service[39] and re-express them in terms of ACS SOC codes.

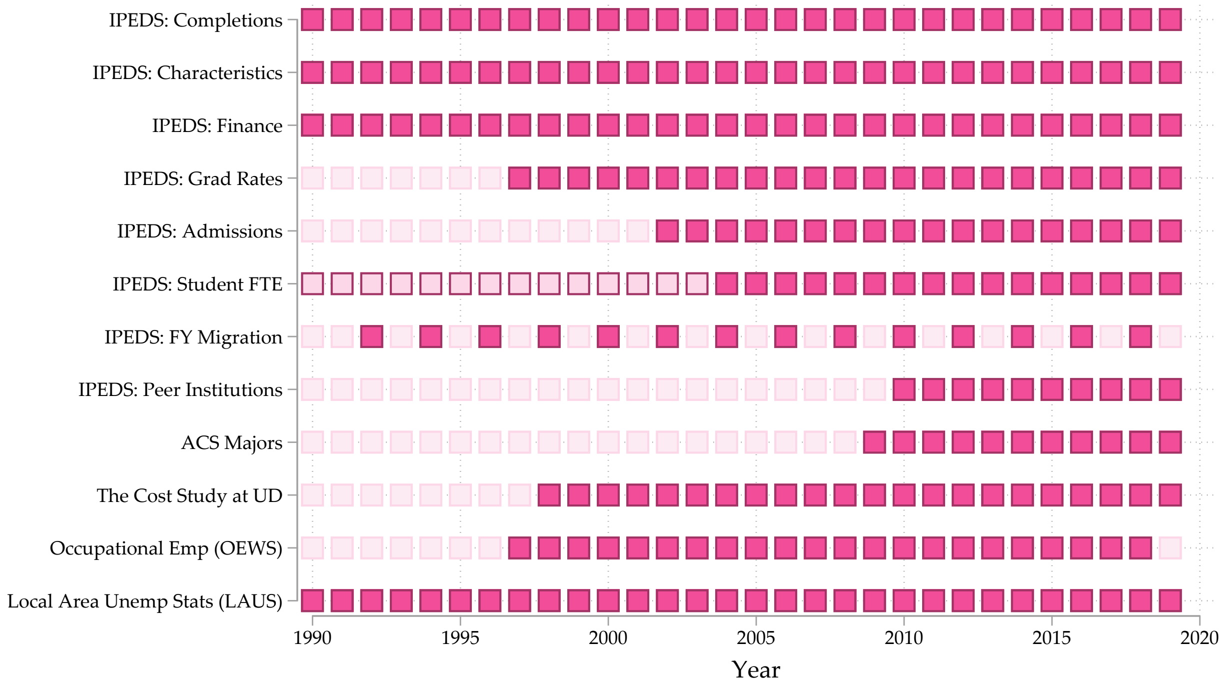
From this harmonized employment series I construct two distinct school-level demand proxies used in the alternative-explanation analyses in Appendix C. The first ("ACS") links OEWS occupations to ACS occupation codes and aggregates occupational employment to the school level using base-year migration shares, without mapping employment to specific majors; it proxies overall demand for bachelor's-level labor. The second ("O*NET") uses O*NET's CIP-to-SOC crosswalk to assign each occupation's employment to the college majors it maps to—dividing employment equally across all related fields—before aggregating to the school level via the same migration share weights. Each approach produces an "all-employment" variant and a "bachelor's-weighted" variant; the latter scales occupational employment by the ACS-estimated share of workers in each occupation holding a bachelor's degree or higher (computed in five-year bins for the ACS approach) or by O*NET education requirements (for the O*NET approach).

For the program-level peer pressure analyses in Appendix D, I also construct a shift-share demand proxy using ACS occupation-to-major mappings. Specifically, I use pooled ACS microdata to estimate the share of workers in each occupation who reported earning a bachelor's degree in a given field, then interact these weights with OEWS state-level employment to form a school-

[39] https://widcenter.org/document/occcodes/

by-major demand index. Because ACS respondents only report college major starting in 2009, the occupation-to-major weights pool the 2009–2019 ACS under the assumption that the mappings are sufficiently stable across broad fields. I describe further how all resultant datasets are used in the next appendix sections.

Figure B1: Data source availability matrix, 1990-2019



Notes: Student FTE was imputed for years 1990 through 2003 counting full time students as 1 and part-time students as half. See text for further details.

Table B2: Descriptive statistics of TCS samples and full IPEDS sample institutions

	Full IPEDS Sample		TCS Perennial		TCS Other	
	Mean	SD	Mean	SD	Mean	SD
Avg Log(EMI)	2.49	0.66	2.89	0.30	2.64	0.47
Control: Public	0.65	0.48	0.95	0.22	0.74	0.44
Control: Private nonprofit	0.32	0.46	0.05	0.22	0.26	0.44
Control: For-profit	0.04	0.19	-	-	-	-
Selectivity: High	0.20	0.40	0.09	0.29	0.15	0.36
Selectivity: Moderate	0.62	0.49	0.85	0.36	0.73	0.44
Selectivity: Low	0.19	0.39	0.06	0.25	0.12	0.32
Highest degree offering: Bachelor's	0.07	0.60	0.00	0.25	0.03	0.48
Highest degree offering: Master's	0.19	0.39	0.05	0.22	0.15	0.36
Highest degree offering: Doctorate	0.74	0.44	0.94	0.23	0.82	0.39
Share BA's awarded: Women	0.57	0.11	0.56	0.06	0.57	0.09
Share BA's awarded: Black	0.09	0.13	0.08	0.10	0.08	0.12
Share BA's awarded: Hispanic	0.08	0.11	0.08	0.11	0.07	0.11
Share BA's awarded: Foreign/Intl.	0.04	0.04	0.02	0.02	0.03	0.04
Share BA's (1990-2019)	1.00		0.16		0.39	
Number of Institutions	3,236		97		524	

Notes: TCS=The Cost Study at the University of Delaware. The perennial sample includes institutions in the main IPEDS analytic sample that participated 10 or more consecutive years in TCS between 1998 and 2019. Other institutions in TCS participated fewer than 10 consecutive years but were also part of the main IPEDS analytic sample.

C. Alternative explanations for major diversification

C.1. Changes in the business cycle and employer demand

An emerging literature ties major choice and colleges' prioritization of certain fields to the business cycle and to employer demand for different skills (e.g., Blom et al., 2021; Conzelmann et al., 2023; Weinstein, 2022; Acton, 2021). If colleges diversified because labor demand for the fields they could offer was rising, that response, rather than peer competition, could account for the trend. I test this in two steps: a business-cycle proxy, and a battery of occupational-employment proxies built from OEWS data under alternative mapping and weighting choices. Throughout, each proxy is entered one at a time alongside peers' average EMI, so that the question is whether the proxy carries any independent relationship with diversification once peer behavior is accounted for.

Effective unemployment. I first construct an effective unemployment rate faced by each college in a given year from time-varying state unemployment rates and the base-year share of the college's incoming first-year students drawn from each state. Base-year (rather than time-varying) shares are standard in shift-share designs (e.g., Chakrabarti et al., 2020) and support an argument for exogeneity of the shares themselves (Goldsmith-Pinkham et al., 2020). The measure is

$$\widehat{EffUnemp}_{i,t} = \sum_{s} \frac{FY_{i,s,t_0}}{FY_{i,t_0}} \cdot \ln(Unemp_{s,t}) \,,$$

for school i and year t, where $FY_{i,s,t_0}/FY_{i,t_0}$ is the share of i's first-year students originating from state s in its base year t_0.

Occupational employment proxies. The unemployment rate captures aggregate conditions but not demand for the specific fields a college might add. To get at field-specific demand I use OEWS state-by-occupation employment for 1997–2018 and connect occupations to a school's potential majors in two ways, each in turn under two employment bases.

Crosswalk. The first mapping ("ACS") links OEWS occupations to ACS occupation codes and aggregates to the school level, remaining agnostic about major; it proxies the simple idea that colleges diversified because overall demand for bachelor's-level labor was rising. The second mapping ("O*NET") links occupations to majors through a CIP-to-SOC crosswalk, dividing each occupation's employment equally across the majors it maps to before aggregating; it ties demand more tightly to the fields a school could plausibly offer.

Employment base. Under each crosswalk I build an "all-employment" version and a "bachelor's" version. The latter scales occupational employment by the share of that occupation held by workers with a bachelor's degree or higher—taken from the ACS (in five-year bins) for the ACS mapping and from O*NET for the O*NET mapping—isolating demand for college-level work within each occupation.

Each of the four resulting series is re-weighted to the school level by the same base-year migration shares used for effective unemployment, $\sum_s (FY_{i,s,t_0}/FY_{i,t_0}) \cdot \ln(\cdot)$, yielding a school-by-year exposure to occupational demand.

Effective occupation index. Finally, I summarize not the level of demand but its *dispersion*. Mirroring the construction of the EMI, I compute an inverse-Herfindahl index of bachelor's-level employment across occupations, re-weighted to each school by its base-year migration shares:

$$\widehat{EOI}_{i,t} = \left(\sum_o \left[\frac{\widetilde{Emp}_{o,i,t}}{\sum_{o'} \widetilde{Emp}_{o',i,t}} \right]^2 \right)^{-1},$$

where $\widetilde{Emp}_{o,i,t}$ is school i's migration-weighted exposure to employment in occupation o. A higher EOI means the labor demand a school faces is spread across more occupations—a demand-side analog of the diversification it might supply.

Results. Table C1 reports each proxy entered on its own (Panel A), then with peers' average EMI added under OLS (Panel B) and under the two instruments used in the main text (Panels C and D). Two patterns hold across the table. First, the peer effect is large, positive, and precisely estimated in every specification, with instrumented elasticities in the same range as the main estimates; it is unmoved by which demand control is included.[40] Second, none of the four occupational-demand *levels* carries an independent relationship with the EMI: the coefficients are small, statistically indistinguishable from zero, and not systematically signed, whether or not peers enter and whether estimated by OLS or instrumental variables. Effective unemployment is similarly weak, with an elasticity near 0.03–0.05 that is at most marginally significant.

The one occupational measure with any signal is the effective occupation index, which enters

[40] The instrumented peer elasticities here ($\approx$ 2.8–3.3 in Panel C) run somewhat above the headline estimate of 1.98 because the occupational-demand data restrict the sample to roughly 32,000 school-years with non-missing OEWS coverage rather than the full ~43,000; on that same restricted sample the main specification produces a comparably elevated estimate. The business-cycle column, which retains the full sample, reproduces the 1.98 estimate.

negatively and is significant at the five percent level under OLS (Panels A and B). The sign is not what a demand-driven account of diversification would predict—more diffuse occupational demand is associated with *less*, not more, major diversification—and the relationship attenuates toward zero once peers' EMI is instrumented: it is insignificant under the $Y_{k(i)}$ instrument (Panel C) and only marginally significant (ten percent) under $X_{k(i)}$ (Panel D). Read together, the levels show no demand channel, and the only dispersion measure that does correlate with the EMI both points the wrong way for a demand story and weakens substantially under the peer instruments. Neither displaces the peer effect.

Long-difference robustness. A level proxy entered under unit and year fixed effects is identified off a school's deviations from its own mean; it does not ask whether a school that experienced rising demand diversified more. Because the migration shares are fixed at the base year, first-differencing the proxy yields a clean shift-share growth term—base-year shares times occupation-employment growth—so the same series answers the growth question once both sides of the equation are differenced. Table C2 reports six-year long differences, the horizon used for the long-difference robustness elsewhere in the paper, estimated by 2SLS with the differenced excluded peers-of-peers instrument. The longer horizon is also the better-identified of the differenced options: a six-year change carries more of the slowly-propagating peer signal than a one- or three-year change, so its first stage is stronger and its estimates less attenuated. The conclusion is unchanged. None of the four occupational-demand growth terms carries an independent relationship with diversification, the business-cycle growth term is likewise null, and the instrumented peer effect remains large and precise.[41] One caveat carries over from the levels: the year effects survive differencing and absorb the common, sector-wide demand movement, so this exercise—like the levels—identifies only *differential* demand growth across schools, not the aggregate demand trend. Whether the rising demand tide lifted sector-wide diversification is a separate, descriptive question that these within-year comparisons cannot address.

C.2. Graduate Education Spillovers

To understand possible effects from graduate program spillovers on the undergraduate EMI, I first create an analogous graduate EMI for the combined master's and doctoral degrees awarded by each institution and year as well as the (log) count of all graduate degrees awarded in a year

[41] The effective occupation index is omitted here: it is a dispersion index rather than a demand flow, so a differenced version has no clean interpretation. The not-offered/offered ratio is omitted throughout for the reason given above.

Table C1: Employer Demand Proxies and Major Diversification

	(1) Bus. cycle	(2) OEWS occ. demand	(3)	(4)	(5)	(6) Dispersion
	Eff. unemp.	All occ O*NET	All occ ACS	BA occ O*NET	BA occ ACS	Occ. index
Panel A. Demand proxy only (OLS), Outcome = Log(EMI)						
Log(demand proxy)	0.022	−0.089	−0.139	0.050	−0.035	−0.270*
	(0.023)	(0.151)	(0.164)	(0.109)	(0.137)	(0.107)
Panel B. Adding peer EMI (OLS), Outcome = Log(EMI)						
Log(demand proxy)	0.028	−0.053	−0.097	0.072	−0.006	−0.254*
	(0.022)	(0.141)	(0.153)	(0.106)	(0.132)	(0.105)
Log(Peers' Avg. EMI_t)	0.312**	0.257**	0.256**	0.259**	0.258**	0.235**
	(0.077)	(0.084)	(0.084)	(0.083)	(0.083)	(0.082)
Panel C. Adding peer EMI (IV, instrument $\overline{Y}_k$), Outcome = Log(EMI)						
Log(demand proxy)	0.050	0.140	0.140	0.146	0.148	−0.104
	(0.026)	(0.172)	(0.177)	(0.144)	(0.166)	(0.139)
Log(Peers' Avg. EMI_t)	1.979**	2.810**	2.813**	2.797**	2.806**	3.283**
	(0.291)	(0.686)	(0.687)	(0.679)	(0.683)	(1.179)
First-stage F	109.9	27.2	27.2	27.6	27.5	11.3
Panel D. Adding peer EMI (IV, instrument $\overline{X}_k$), Outcome = Log(EMI)						
Log(demand proxy)	0.029	0.039	0.038	0.080	0.053	−0.088
	(0.023)	(0.121)	(0.127)	(0.101)	(0.117)	(0.084)
Log(Peers' Avg. EMI_t)	1.216**	1.389**	1.390**	1.384**	1.390**	0.882**
	(0.269)	(0.338)	(0.338)	(0.339)	(0.338)	(0.325)
First-stage F	22.4	18.6	18.5	18.6	18.8	18.2
School FEs	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓
N Observations	42,903	32,638	32,638	32,638	32,638	30,201
N Clusters	1,690	1,679	1,679	1,679	1,679	1,688

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All models include institution and year fixed effects and are weighted by the base-year total number of BA degrees awarded. Each column enters a single demand proxy: (1) effective unemployment from state unemployment rates and base-year migration shares; (2)–(5) migration-weighted occupational employment, under the O*NET/CIP-SOC or ACS occupation-to-major crosswalk and using all employment or bachelor's-level employment only; (6) the effective occupation index, an inverse-Herfindahl of bachelor's-level employment across occupations. Panel A enters the proxy alone; Panels B–D add peers' average EMI, estimated by OLS (B) and by instrumental variables using the excluded peers-of-peers EMI, $Y_{k(i)}$ (C), or excluded-peer control averages, $X_{k(i)}$ (D), as in the main text. Sample sizes shown are for the Panel C specification; the OLS columns use marginally more observations. Instrumenting each demand proxy with its own lagged value (to address measurement error) leaves the demand coefficients substantively unchanged and is omitted for space. The not-offered/offered relative-demand ratio is excluded because its construction is mechanically tied to the number of fields a school offers, and hence to the EMI itself.

Table C2: Employer Demand and Major Diversification: Six-Year Long Differences

	(1)	(2)	(3)	(4)	(5)
	Bus. cycle	OEWS occupational demand			
	Eff. unemp.	All occ O*NET	All occ ACS	BA occ O*NET	BA occ ACS
Outcome = Δ_6 Log(EMI); 2SLS, instrument $\overline{Y}_k$					
Δ_6 Log(demand proxy)	0.016	0.125	0.127	0.069	0.072
	(0.022)	(0.184)	(0.174)	(0.168)	(0.180)
Δ_6 Log(Peers' Avg. EMI)	2.382**	2.840**	2.842**	2.827**	2.832**
	(0.458)	(0.897)	(0.900)	(0.888)	(0.892)
First-stage F	41.5	12.8	12.8	12.9	12.8
Base-year FEs	✓	✓	✓	✓	✓
N Observations	32,880	22,697	22,697	22,697	22,697
N Clusters	1,602	1,595	1,595	1,595	1,595

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. Each variable is a six-year long difference, $\Delta_6 x_{it} = x_{it} - x_{i,t-6}$; differencing removes the unit fixed effect, while year effects are retained. All models are weighted by the base-year (i.e., $t-6$) total number of BA degrees awarded and estimated by 2SLS, instrumenting the differenced peer EMI with the differenced excluded peers-of-peers EMI, $Y_{k(i)}$, as in the main text. Column (1) is the effective unemployment rate (business cycle); columns (2)–(5) are the migration-weighted occupational-demand levels from Table C1, differenced. The effective occupation index and the not-offered/offered ratio are omitted (see text).

from IPEDS. I report a series of regression results to test whether indicators of these graduate degree diversity measures are related to the undergraduate EMI. Because schools need to be awarding graduate degrees to have a value for the EMI and total degrees, this is a test of the intensive margin of graduate spillovers. In each specification, I control for demographic and enrollment characteristics of the focal institution, and add various peer measures to test sensitivity of their relationships to the EMI and the main results in section 5.

The results in Table C3 show an elasticity of about 0.1 between the graduate and undergraduate EMI. That is, shown in Panel A, a 10 percent increase in the graduate EMI was associated with a 1 percent increase in undergraduate EMI. In Panel B, the (log) growth in graduate degrees at an institution had a much smaller effect on the undergraduate EMI. This suggests that diversity of graduate degrees, not just increases in their number, is more highly correlated to the undergraduate EMI. This is consistent with the small effects of an institution's enrollment growth on the EMI discussed in the main text.

In either measurement, peer effect estimates are still significantly larger and aligned with the models from the main text that do not control or account for graduate degrees. The patterns seen here cannot definitively rule out simultaneity or reverse causality, whereby undergraduate major diversification causes graduate program diversification, or that the institution strategically diversified its graduate and undergraduate degrees contemporaneously. This does not rule out spillover effects from graduate education, but without credible exogenous variation to move either the graduate or the undergraduate major diversity independently, it is difficult to identify such effects. Yet, from this analysis, it seems unlikely graduate program spillovers would provide a stronger mechanism than peer effects, if they exist.

I also examine the extensive margin of graduate education and its relationship to the undergraduate EMI. If spillovers exist between the two levels, then I would expect schools who expand into graduate education to have larger or faster rates of undergraduate major diversification than schools that remain concentrated on delivering bachelor's degrees. To test this, I track the highest degree offerings of each school over time and identify switching institutions that expanded from baccalaureate education into master's or doctoral programs. Schools become treated if they begin offering a master's or doctoral degree of any type, including professional practice degrees (e.g., law). Institutions that only ever offer bachelor's degrees or lower serve as the comparison group ("stayers").

I start with a dynamic Two-Way Fixed Effects (TWFE) specification,

$$\text{Log}(\text{EMI}_{i,t}) = \sum_{\tau \neq -1} \beta_{\tau}^{es} \left(\text{Treated}_i 1\{t = \tau\}\right) + \mathbf{\Gamma X}_{i,t} + \delta_i + \delta_t + \epsilon_{i,t}, \tag{C1}$$

where each β_{τ}^{es} yields an estimate of the difference between treated and untreated units at that time relative to when "switchers" first added graduate degrees. In this setting, there is significant variation in the timing of "treatment" given the length of the panel and lack of an event that would induce institutions to begin offering graduate degrees at the same time. To address differential treatment timing, I also estimate event studies as proposed by Sun & Abraham (2021) and Callaway & Sant'Anna (2021).

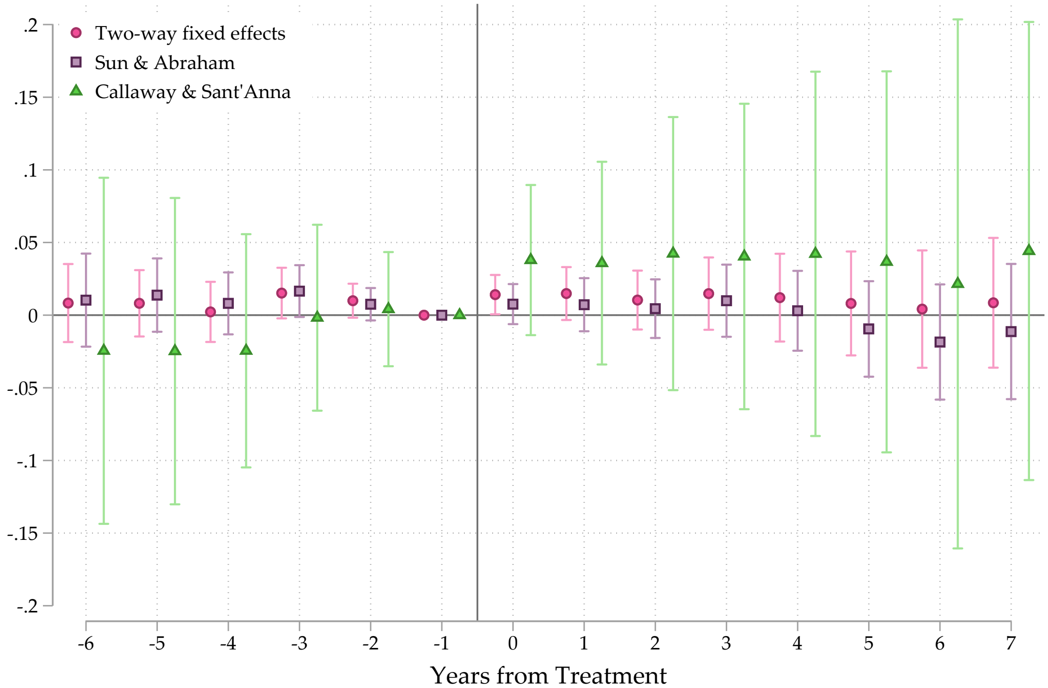
I plot relative-treatment timing estimates for pre- and post-treatment periods for all three estimators in Figure C1. Regardless of the estimator, the main conclusion is the same. Adding graduate education does not lead to increases in the bachelor's degree EMI compared to bachelor's degree institutions who did not add the capacity for graduate programs. Even 7 years post-treatment, the EMI remains roughly the same as "stayers." The Callaway & Sant'Anna (2021) specification yield positive though increasingly imprecise estimates over time, none of which exclude zero in the 95 percent confidence interval.

Table C3: Graduate Degrees and Major Diversification

	(1) OLS	(2) OLS	(3) 2SLS	(4) GMM 2S
Panel A. Graduate Degree Diversity, Outcome = Log(EMI)				
Log(Graduate EMI)	0.122**	0.113**	0.095**	0.099**
	(0.019)	(0.018)	(0.019)	(0.016)
Log(Peers' Average EMI$_t$)		0.258**	2.051**	1.128**
		(0.092)	(0.368)	(0.268)
Instrument(s)	-	-	$\overline{Y}_k$	$\overline{X}_k$
First-stage F	-	-	64.4	22.0
School FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Peer controls		✓	✓	✓
N Observations	31,126	31,120	31,019	31,019
N Clusters	1,301	1,301	1,296	1,296
Panel B. Graduate Degree Awards, Outcome = Log(EMI)				
Log(Graduate Degrees)	0.029*	0.018	-0.010	0.005
	(0.012)	(0.012)	(0.014)	(0.012)
Log(Peers' Average EMI$_t$)		0.271**	2.179**	1.281**
		(0.093)	(0.395)	(0.294)
Instrument(s)	-	-	$\overline{Y}_k$	$\overline{X}_k$
First-stage F	-	-	62.9	21.6
School FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Peer controls		✓	✓	✓
N Observations	31,126	31,120	31,019	31,019
N Clusters	1,301	1,301	1,296	1,296

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All models are weighted by the base-year total number of BA degrees awarded. Panel A presents estimates for the graduate degree EMI on the log undergraduate EMI. Panel B is interested in the relationship between the log number of graduate degrees awarded on the log undergraduate EMI. All models control for logged undergraduate enrollment, share of a school's total enrollment that is part-time, share graduate students, the share of bachelor's degrees awarded to women, to students that were either Black or Hispanic, to foreign or international students, and the number of years in the panel the school had awarded BA degrees and its quadratic. Columns (2)–(4) add the same controls averaged across the institution's peers. Columns (3) and (4) instrument for peers' average EMI using the average EMI of excluded peers, $\overline{Y}_{k(i)}$, and the average control values of excluded peers, $\overline{X}_{k(i)}$, respectively, as in the main text.

Figure C1: Event Study Estimates of Effect of Graduate Program Introduction on Log(EMI)



Notes: Timing is relative to the first year in which institutions began offering graduate degrees. Comparison institutions are those who offered a bachelor's degree (and lower) throughout the panel. Estimates are weighted by BA degrees awarded in each year.

While the analyses of graduate education spillovers do not completely rule out a role for them in explaining the undergraduate EMI they do suggest this is not likely a main driver. Schools that added capacity for graduate programs or grew their existing graduate programs did experience larger increases in the EMI, but this could simply reflect overall institutional priorities for diversification of programs, not that graduate education caused diversification in undergraduate majors. More concretely, expanding into graduate education did not lead to increases in the undergraduate EMI compared to institutions remaining committed to baccalaureate education.

C.3. Declining state support

The Oaxaca-Blinder decomposition in Section 4.2 suggested some role for state support in major diversification, particularly among less selective public institutions. I test whether the share of total non-hospital revenue generated from state appropriations and financial aid to students or the (log) level of this state support per FTE had an effect on changes in the EMI. The level of appropriations is more commonly used in the literature to test effects of state support on outcomes, though in the case of major diversity, it is plausible that the share of revenue generated from state sources might matter as well to institutional decisions. Particularly public non-selective institutions, who tend to rely more heavily on state support, but whose levels of support per FTE may not reflect this, could make curricular decisions based on changes to these shares.

The results in Table C4 show a positive relationship between state support and major diversification, albeit small in the aggregate. Schools receiving support from the state in the form of appropriations or financial aid awarded to students saw a decline in their EMI by 0.1 to 0.15 percent for every 10 percent decline in state support.[42] These relationships among the full sample of institutions shrink and become imprecise when accounting for peer effects.

The lack of an average relationship fails to capture the importance of state support for moderately and less selective public institutions (e.g., regional public universities). Among public institutions that were classified below the top two Barron's competitiveness categories (Most and Highly competitive), the relationship between state support and major diversification is roughly four times larger. A 10 percent decline in state support led to a 0.5 percent decline in major diversity. This relationship is relatively stable even after adding peer effects. Worth noting is the point estimate in the IV peer effects model for this group of institutions is smaller than for the full sample, at about 1.1 to 1.5 depending on the instrument. Though strong, this suggests declining state support may have limited the extent to which these schools could improve their educational offerings in response to investments being made by institutions in other parts of their peer networks.

C.4. Robustness of peer effects to competing explanations

The preceding sections examined each alternative explanation—employer demand, graduate program expansion, and declining state appropriations—on its own. This section enters them together, asking whether the peer effect survives once the leading rivals are controlled simultaneously, at both levels of the analysis. The one asymmetry is that state appropriations vary at the school level and have no field-level analog, so they enter the school-level horse-race but not the program-level one.

Table C5 reports the school-level specification: log EMI on peers' average log EMI, alongside graduate EMI, log state appropriations per FTE, and the bachelor's occupational-demand proxy, with coefficients interpretable as elasticities. The peer effect is unchanged from its baseline. In each case it is large, positive, and precise under OLS, 2SLS, and GMM 2S, with instrumented estimates in

[42] The majority of state support goes to public institutions, however, over 70 percent of private institutions and even 56 percent of for-profit institutions receive some form of state aid. The amounts per FTE are substantially smaller at non-public institutions. Public colleges received over 12 thousand dollars per FTE, while private nonprofit and for-profits received 920 and 474 dollars per FTE on average, respectively (2019 dollars).

Table C4: State Support and Major Diversification

	Full Sample			Public, Less Selective		
	(1) OLS	(2) 2SLS	(3) GMM 2S	(4) OLS	(5) 2SLS	(6) GMM 2S
Panel A. State share of total revenue, Outcome = Log(EMI)						
Log(State Share$_t$)	0.014*	0.000	0.003	0.056**	0.034*	0.032*
	(0.005)	(0.006)	(0.005)	(0.017)	(0.016)	(0.015)
Log(Peers' Average EMI$_t$)		2.012**	1.292**		1.873**	1.301**
		(0.319)	(0.310)		(0.387)	(0.356)
Instrument(s)	-	$\overline{Y}_k$	$\overline{X}_k$	-	$\overline{Y}_k$	$\overline{X}_k$
First-stage F	-	98.7	21.1	-	62.9	17.0
School FEs	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓
Peer controls		✓	✓		✓	✓
N Observations	35,867	35,728	35,728	12,902	12,854	12,854
N Clusters	1,571	1,560	1,560	489	485	485
Panel B. State appropriations and aid per FTE, Outcome = Log(EMI)						
Log(State support per FTE$_t$)	0.013*	0.001	0.005	0.057**	0.039*	0.035*
	(0.006)	(0.007)	(0.006)	(0.017)	(0.017)	(0.016)
Log(Peers' Average EMI$_t$)		2.012**	1.289**		1.879**	1.323**
		(0.317)	(0.309)		(0.390)	(0.359)
Instrument(s)	-	$\overline{Y}_k$	$\overline{X}_k$	-	$\overline{Y}_k$	$\overline{X}_k$
First-stage F	-	99.3	21.2	-	62.3	16.9
School FEs	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓
Peer controls		✓	✓		✓	✓
N Observations	35,977	35,836	35,836	12,969	12,919	12,919
N Clusters	1,576	1,564	1,564	494	489	489

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All models are weighted by the base-year total number of BA degrees awarded. Columns (1)–(3) present estimates on the full sample, while columns (4)–(6) are run only on public institutions outside the top 2 most competitive categories by Barron's measure (most and highly competitive). Panel A presents estimates for the log share of total revenue from state appropriations and aid on the log EMI. Panel B is interested in the relationship between the (level) log of state appropriations and aid per FTE on the log EMI. All models control for logged undergraduate enrollment, share of a school's total enrollment that is part-time, share graduate students, the share of bachelor's degrees awarded to women, to students that were either Black or Hispanic, to foreign or international students, and the number of years in the panel the school had awarded BA degrees and its quadratic. Columns (2)–(3) and (5)–(6) add the same controls averaged across the institution's peers. Columns (2) and (5) instrument for peers' average EMI using the average EMI of excluded peers, $\overline{Y}_{k(i)}$, and columns (3) and (6) use the average control values of excluded peers, $\overline{X}_{k(i)}$, as in the main text.

the range of the headline results. Among the rivals, only graduate EMI carries a robust association, an elasticity near 0.08, an order of magnitude below the peer effect; state appropriations are small and only intermittently significant, and employer demand is null throughout. Entering all three at once does not move the peer effect. This comparison should be read with one asymmetry in mind: the peer effect is identified using an instrument, while graduate EMI, state appropriations, and employer demand enter without an analogous correction for their own endogeneity, so their smaller and noisier coefficients may partly reflect this difference in identification strategy rather than genuinely weaker effects.

Table C6 repeats the exercise for program adoption. Here the treatments are standardized, so each coefficient is the effect of a one-standard-deviation increase; the columns pair the two peer-adoption measures with the occupation-to-major demand shift-share and field-level master's and doctoral degree counts. The pattern matches: the peer measures dominate and grow under the peers-of-peers instrument, master's degree counts retain a small stable association, and the demand shift-share is null. As with diversification, no competing explanation displaces the peer effect on adoption.

Table C5: Peer Effects Net of Competing Explanations: School Level

	(1) OLS	(2) 2SLS	(3) GMM 2S
Outcome = Log(EMI); treatments lagged one year			
Peers' avg. Log(EMI)	0.341**	2.641**	1.198**
	(0.083)	(0.794)	(0.369)
Graduate EMI	0.097**	0.077**	0.082**
	(0.019)	(0.021)	(0.017)
State approp. / FTE	0.014*	0.004	0.012*
	(0.007)	(0.010)	(0.006)
BA occ. demand	−0.053	−0.002	0.004
	(0.115)	(0.136)	(0.109)
Instrument	-	$\overline{Y}_k$	$\overline{X}_k$
First-stage F	-	17.0	17.0
School FEs	✓	✓	✓
Year FEs	✓	✓	✓
N Observations	21,057	21,014	21,014
N Clusters	1,207	1,204	1,204

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. All variables are in logs, so coefficients are elasticities; all are lagged one year. Models include institution and year fixed effects and are weighted by the base-year total number of BA degrees awarded. Columns (2)–(3) instrument peers' average log EMI with the excluded peers-of-peers EMI, $Y_{k(i)}$ (2SLS), or the excluded-peer control averages, $X_{k(i)}$ (GMM 2S), as in the main text; the reported F is the first-stage statistic. Graduate EMI is the inverse-HHI diversification index computed over graduate degrees; state appropriations are real (CPI-adjusted) state and local appropriations per FTE; BA occupational demand is the migration-weighted bachelor's occupational-employment proxy of Table C1.

Table C6: Peer Effects Net of Competing Explanations: Program Adoption

	N peers with program		Peer degree awards	
	(1) OLS	(2) 2SLS	(3) OLS	(4) 2SLS
Outcome = 1(first year field offered); standardized, $t-1$				
N Peers with program	0.0175**	0.0442**		
	(0.0011)	(0.0034)		
Peer degree awards (100s)			0.0080**	0.0165**
			(0.0013)	(0.0019)
Occ. shift-share	0.0007	0.0002	0.0003	−0.0004
	(0.0011)	(0.0011)	(0.0011)	(0.0011)
N Master's degrees	0.0008**	0.0008**	0.0008**	0.0008**
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
N Doctoral degrees	0.0013	0.0012	0.0012	0.0011
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
First-stage F	-	823.8	-	320.2
Major × year FEs	✓	✓	✓	✓
School × major FEs	✓	✓	✓	✓
N (school-field-years)	1.41m	1.40m	1.41m	1.40m
N schools	1,692	1,681	1,692	1,681
Outcome mean	0.007	0.007	0.007	0.007

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. The outcome equals one in the first year a school awards BA degrees in a field, conditional on not having offered it previously. All treatments are standardized to unit variance, so each coefficient is the effect of a one-standard-deviation increase, and are lagged one year. Every column is a horse-race entering a peer-adoption measure together with the occupation-to-major demand shift-share and the field's master's and doctoral degree counts; columns (1)–(2) use the count of peers offering the field, columns (3)–(4) total peer awards in the field. Columns (2) and (4) instrument the peer measure with the corresponding peers-of-peers measure; the reported F is the first-stage statistic. All specifications include major-by-year and school-by-major fixed effects. State appropriations are omitted: they vary at the school level and have no field-level analog.

D. TCS Department-Level Cost Analyses

This section provides additional details on the department-level analyses of program introduction costs presented in Section 6. All analyses in this section use the perennial TCS sample of 97 institutions described in Appendix B.

Event identification

I identify new bachelor's degree program introductions at perennial TCS institutions using IPEDS degree completion records. A program introduction occurs at institution i in field m if the first year of positive degree completions in that field is preceded by at least five consecutive years of zero completions. This lookback requirement avoids misclassifying programs that temporarily stopped awarding degrees as new introductions. I impute the actual year of program origination as one year before the first year of degree completions, reflecting that departments typically begin enrolling students in new coursework before the first cohort completes a degree. I further require that at least five students completed the program in the first year of positive completions. The introduction events span 2002 through 2017, reflecting both the lookback requirement and the end of the TCS panel.

Cost ratio at introduction

For the cost ratio analysis presented in Figure 6 and Table D1, I retain all introduction events for which at least five consecutive years of post-introduction TCS data are available (years 0 through 4 relative to the imputed introduction year). This yields 198 introduction events. I then compute a cost ratio for each event at each post-introduction year as follows.

I first construct real instructional costs per credit hour for each institution-field-year cell in the TCS panel. Direct instructional expenditures are deflated to 2019 dollars using the Consumer Price Index for All Urban Consumers (CPI-U) from the Federal Reserve Economic Data (FRED) series and divided by student credit hours taught, net of upper-division instructional and graduate instructional credit hours.

The cost ratio is defined as a *benchmark* divided by the introducing institution's *portfolio* average. The benchmark for a given introduction event (institution i, field m, year t) is the leave-one-out mean cost per credit hour across all other perennial TCS institutions that are "established" in field m at time t, where established means having at least three consecutive prior years of positive credit hours in that field. This leave-one-out construction ensures that institution i's own costs do not enter its benchmark. The portfolio is the mean cost per credit hour across all other programs that institution i is already operating in year t, excluding the newly introduced field. A ratio above one indicates the new program type is more expensive per credit hour at institutions where it is already established than the introducing school currently spends on average across its existing programs.

Of the 198 events, 162 have non-missing cost ratios at the introduction year. Table D1 reports the number of introductions and mean cost ratios by broad field.

Table D1: New Program Cost Ratios by Broad Field at Introduction

Field	N Intros	Avg Benchmark ($/CH)	Avg Portfolio ($/CH)	Avg Cost Ratio†
Engineering	30	2,001.47	752.75	2.848
Physical Sciences	8	2,058.02	706.18	2.542
Humanities	12	1,073.72	596.35	1.931
Education	8	873.65	499.19	1.759
Agriculture	7	935.11	686.93	1.439
Health	47	904.06	735.76	1.411
Business	15	571.62	552.01	1.106
Social Sciences	17	732.80	719.52	1.098
Arts	4	760.34	800.13	1.031
Communications	14	426.90	665.33	0.668
Overall weighted avg	162	1,083.14	690.11	1.655

Notes: Each row reports the mean cost ratio across introduction events in the given broad field at the imputed year of introduction ($\tau = 0$). The cost ratio is the benchmark (leave-one-out mean cost per credit hour across other established TCS institutions in the same field) divided by the portfolio (mean cost per credit hour across the introducing institution's other existing programs). The overall weighted average weights field-level means by the number of introductions in each field. Cost per credit hour is in 2019 dollars. See text for details.
† Avg Cost Ratio is the mean of each event's own benchmark-to-portfolio ratio, not the ratio of the Avg Benchmark and Avg Portfolio columns shown; these will generally differ whenever portfolio costs vary across events within a field.

Stacked event study design

To estimate the effect of program introductions on costs and enrollment at existing ("bystander") departments, I use a stacked difference-in-differences design following Cengiz et al. (2019). This approach addresses concerns about bias from heterogeneous treatment timing in conventional TWFE event studies (e.g., de Chaisemartin & D'Haultfœuille, 2020; Goodman-Bacon, 2021) by constructing a separate sub-experiment for each introduction event and estimating a pooled model with stack-specific fixed effects.

The stacked event study uses a stricter sample than the cost ratio analysis because it requires data coverage for the full event window. Each event requires seven consecutive years of TCS data (three pre-introduction years and three post-introduction years, plus the introduction year itself) for the introducing institution and its comparison units. After applying this requirement, 126 candidate events remain. Each candidate event is then constructed as a "stack" consisting of treated and control department-year observations.

Treated units. For each event at institution i introducing field m, the treated units are all of institution i's other existing departments (i.e., fields other than m) that have complete data across the seven-year event window.

Control units. For each event, the control pool consists of same-field departments at other perennial TCS institutions that also have complete seven-year coverage. Two types of institutions are excluded from the control pool for a given event. First, I exclude institutions identified as directed peers of the introducing school, using peer lists reported to IPEDS. Second, I exclude any institution that itself introduced any new program (by the same lookback-based definition) during the seven-year event window, to avoid contamination from concurrent program introductions. I further require at least three distinct control institutions per stack.

Of the 126 candidate events, 53 satisfy all stacking requirements, yielding 53 final stacks across 62 distinct institutions and 217,546 total observations.

Estimation

The pooled stacked event study is estimated via high-dimensional fixed effects regression using `reghdfe`. For each outcome Y (log cost per credit hour, log instructional expenditures, log credit hours, among others), the estimating equation is

$$Y_{m,i,s,t} = \sum_{\tau \neq -1} \beta_\tau \left(\text{Treat}_{i,s} \times 1\{t = \tau\}\right) + \gamma \cdot \text{GradShare}_{m,i,t} + \alpha_{s,i,m} + \lambda_{s,\tau,m} + \epsilon_{m,i,s,t}, \tag{D1}$$

where m indexes department (field), i indexes institution, s indexes the stack, and τ indexes event time relative to the imputed introduction year. The model includes stack-by-institution-by-department fixed effects ($\alpha_{s,i,m}$) and stack-by-event-time-by-department fixed effects ($\lambda_{s,\tau,m}$). The $\text{Treat}_{i,s}$ indicator equals one for all departments at the introducing institution in stack s and zero for departments at control institutions. I control for the share of credit hours taught at the graduate level, as costs per credit hour in TCS cannot be separated by instructional level. Event time runs from $\tau = -4$

to $\tau = +4$, with the endpoints serving as binned categories for observations outside the core window. The reference period is $\tau = -1$. All regressions are weighted by the pre-period mean of departmental credit hours and standard errors are clustered at the institution level.[43]

Peer pressure measures

To investigate whether the cost consequences of program introduction differ by competitive motivation, I construct a measure of peer pressure for each introduction event. For event (institution i, field m, year t), I identify all directed peers of institution i from the IPEDS peer lists and count the number of distinct programs that those peers introduced in the five years prior to t (years $t-5$ through $t-1$) that institution i did not already offer. I then split events at the median of this count. Events above the median are classified as "high peer pressure" introductions, reflecting cases where the focal institution's peers had been actively expanding into fields the focal institution had not yet entered.

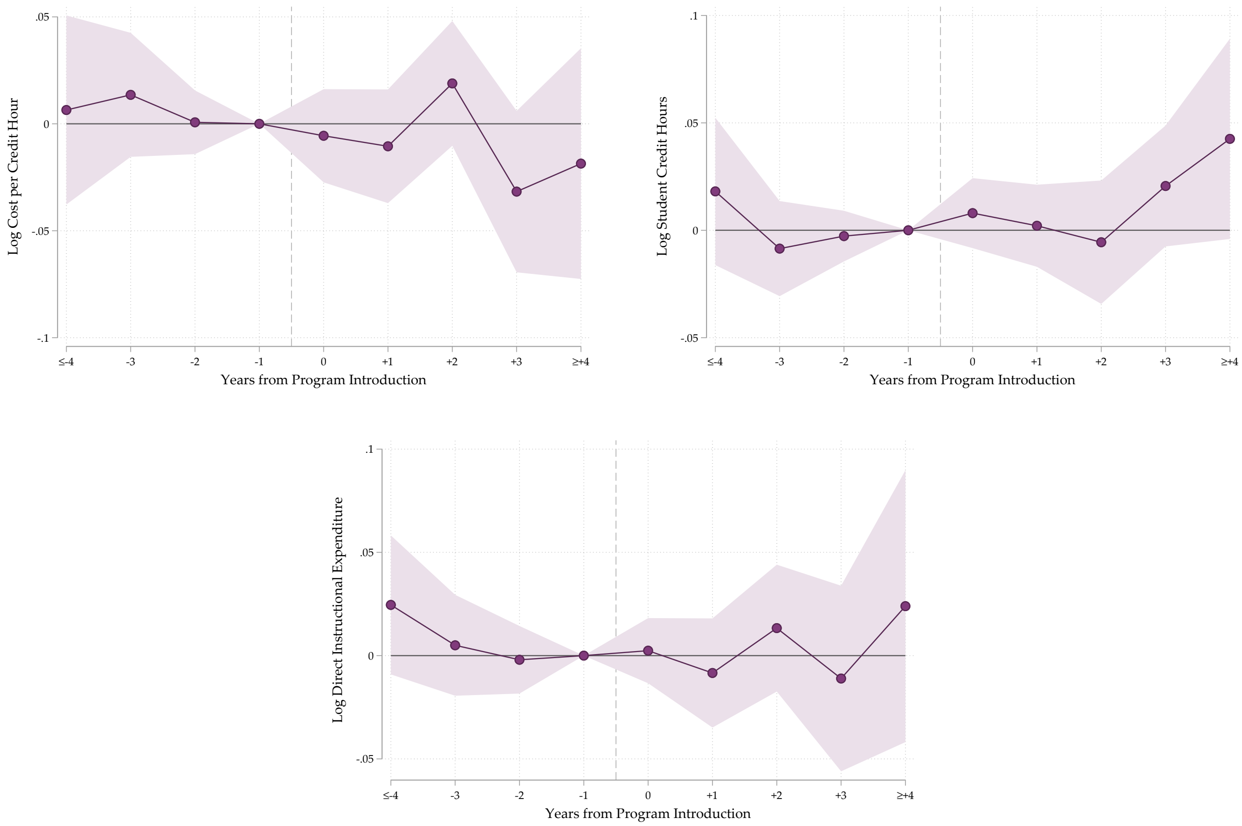
[43]Costs per credit hour cannot be determined separately for graduate and undergraduate credits.

Table D2: Stacked Event Study: Pooled Post-Period Averages, Bystander Departments

	Log Cost per Cr. Hr.	Log Student Cr. Hrs.	Log Direct Inst. Exp.
Post-period avg. ($\tau = 0$ to 4)	-0.010	0.014	0.004
	(0.012)	(0.011)	(0.016)
p-value	0.445	0.238	0.796
Post $F(5, 61)$ p-value	0.034	0.036	0.035
Pre $F(3, 61)$ p-value	0.769	0.233	0.406
N Observations		216,940	
N Clusters		62	

Notes: Post-period average is a linear combination of the $\tau = 0$ through $\tau = 4$ event-time treatment coefficients from equation (D1). Standard errors in parentheses; p-values are two-tailed with 61 degrees of freedom (clusters minus one). Post F tests joint significance of the five post-period coefficients ($\tau \in \{0, 1, 2, 3, 4\}$); the post-average is not significant for any outcome. Pre F tests joint significance of the $\tau \in \{-4, -3, -2\}$ pre-trend coefficients. Regressions weighted by pre-period mean departmental credit hours; standard errors clustered at the institution level.

Figure D1: Stacked Event Study: Bystander Department Outcomes Around New Program Introduction



Notes: Each panel plots pooled stacked event study estimates with 95% confidence intervals for bystander departments at introducing institutions relative to same-field departments at non-introducing TCS control institutions. The event window spans $\tau \in [-3, 4]$ relative to the imputed introduction year; $\tau = -1$ is the omitted reference period. Panels show log cost per credit hour (top left), log student credit hours (top right), and log direct instructional expenditure (bottom center). Regressions include stack-by-institution-by-department and stack-by-event-time-by-department fixed effects, and control for the graduate credit hour share. Standard errors clustered at the institution level. $N = 53$ introduction stacks; 62 institutions.

Table D3: Stacked Event Study: Full Event-Time Coefficients, Bystander Departments

	Log Cost per Cr. Hr.	Log Student Cr. Hrs.	Log Direct Inst. Exp.
$\tau = -4$ (binned)	0.006	0.018	0.025
	(0.023)	(0.018)	(0.017)
$\tau = -3$	0.014	-0.009	0.005
	(0.015)	(0.011)	(0.013)
$\tau = -2$	0.001	-0.003	-0.002
	(0.008)	(0.006)	(0.008)
$\tau = -1$		[reference period]	
$\tau = 0$	-0.006	0.008	0.002
	(0.011)	(0.008)	(0.008)
$\tau = 1$	-0.011	0.002	-0.008
	(0.014)	(0.010)	(0.014)
$\tau = 2$	0.019	-0.006	0.013
	(0.015)	(0.015)	(0.016)
$\tau = 3$	-0.032	0.021	-0.011
	(0.019)	(0.014)	(0.023)
$\tau = 4$ (binned)	-0.019	0.043	0.024
	(0.028)	(0.024)	(0.034)
N Observations		216,940	
N Clusters		62	

Notes: Each cell reports the $\hat{\beta}_\tau$ coefficient from equation (D1) for the indicated outcome and event time. $\tau = -1$ is the omitted reference period; $\tau = -4$ and $\tau = +4$ are binned endpoint categories. No coefficient is statistically significant at the 5% level. Standard errors clustered at the institution level in parentheses. Regressions weighted by pre-period mean departmental credit hours.

Table D4: Peer Pressure and Cost of Program Introduction: Robustness Checks

	1	2	3	4
Panel A: Normalized peer count				
High Peer Pressure	0.601**	0.496*	0.491**	0.530**
	(0.216)	(0.194)	(0.187)	(0.198)
Panel B: All program introductions by peers				
High Peer Pressure	0.278	0.279	0.240	0.249
	(0.198)	(0.205)	(0.204)	(0.206)
Panel C: Three-year peer window				
High Peer Pressure	0.488*	0.385*	0.412*	0.464*
	(0.214)	(0.180)	(0.184)	(0.210)
Event-Year FE	✓	✓	✓	✓
Broad Field FE	✓			
Specific Major FE		✓	✓	✓
Selectivity (3-cat.)			✓	✓
Census Region				✓
N Observations	146			
N Clusters	68			

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. Sample fixed at $N = 146$ events, matching Table 8. Outcome is the cost ratio at introduction. Panel A replaces the binary median-split with the continuous peer-gap count normalized by the introducing school's portfolio size. Panel B extends the peer-gap count to include all program introductions by peers, regardless of whether the focal school already offered those fields; the null in this panel supports the mechanism, as the relevant competitive pressure is from peers entering fields the focal school had not yet entered, not from peer activity generally. Panel C uses a three-year peer window (years $t - 3$ through $t - 1$) in place of the five-year window used in Table 8. Fixed effects follow Table 8 exactly across all panels.

Table D5: Peer Pressure and Cost of Program Introduction: Controlling for Portfolio Size

	1	2	3	4
High Peer Pressure	0.607**	0.467**	0.420**	0.466**
	(0.221)	(0.158)	(0.146)	(0.153)
Portfolio size (N programs)	-0.027*	-0.036**	-0.035**	-0.038**
	(0.013)	(0.011)	(0.012)	(0.012)
Event-Year FE	✓	✓	✓	✓
Broad Field FE	✓			
Specific Major FE		✓	✓	✓
Selectivity (3-cat.)			✓	✓
Census Region				✓
N Observations	146			
N Clusters	68			

Notes: * $p < 0.05$, ** $p < 0.01$. Standard errors clustered at the institution level in parentheses. Sample fixed at $N = 146$ events, matching Table 8. Outcome is the cost ratio at introduction. Portfolio size is the count of distinct fields the introducing institution operated in the year before the introduction. The peer pressure effect is robust to controlling for portfolio size, which enters negatively, consistent with the cost ratio benchmark being more difficult to exceed for schools that already operate broad portfolios.